Architecture Shape Governs QNN Trainability: Jacobian Null Space Growth and Parameter Efficiency

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Abstract

Variational quantum circuits with angle encoding implement truncated Fourier series, and architectures arranging N qubits with L encoding layers each — sharing encoding budget $E = NL$ — generate identical frequency spectra, identical frequency redundancy, and require the same minimum parameter count for coefficient control. Despite this equivalence, trainability varies substantially with architecture shape (N, L) at fixed E. We identify structural rank deficiency of the coefficient matching Jacobian J as the mechanism responsible. For serial single-qubit architectures, we prove $\mathrm{rank}(J) \leq 2L + 1$ regardless of parameter count P, with $\dim(\ker J) \geq P - (2L + 1)$ growing without bound — a phenomenon we term *structural gradient starvation*: a growing fraction of parameters become structurally decoupled from the loss as P increases at fixed L. Parallel architectures avoid this via independent phase trajectories, ensuring $\sigma_{\min}(J^{(\mathrm{par})}) > 0$ generically for $P \leq 2E + 1$, so no parameter lies in $\ker J$. For practitioners, we further show that the two natural routes to increasing parameter count have fundamentally different effects: adding feature map (FM) layers monotonically strengthens the Jacobian QFIM eigenvalue spectrum and achieves $R^2 \geq 0.95$ with 1.6–2.2$\times$ fewer parameters than adding trainable blocks across all tested architectures, while trainable blocks improve training only through the classical interpolation mechanism with no quantum-specific benefit.

1 Introduction

Variational quantum circuits (VQCs) with angle encoding implement truncated Fourier series (Schuld et al., 2021), and are universal function approximators even for a single qubit (Pérez-Salinas et al., 2021). Arranging N qubits with L encoding layers each, the accessible frequency spectrum depends only on the total encoding budget $E = NL$, not on how it is distributed across (N, L) at fixed E (Holzer & Turkalj, 2024). Under unary encoding, frequency redundancy — the spectral bias toward lower frequencies (Rahaman et al., 2019) arising from combinatorial degeneracy in the encoding layers — is likewise governed entirely by E (Duffy & Jastrzebski, 2026). Both the accessible function class and the spectral bias mechanism are therefore architecture-independent. Holzer & Turkalj (2024) observe that larger N tends to yield more favorable optimization behavior, and note that a large frequency spectrum can make accurate training harder even when theoretical expressivity is sufficient. Figure 1 makes this precise: at fixed E — where all architectures share identical spectra and redundancy structure — low-qubit ($N = 1, 2$) architectures fail across the full parameter range while intermediate architectures ($N = 3, 4$) succeed with far fewer parameters. Since single-qubit

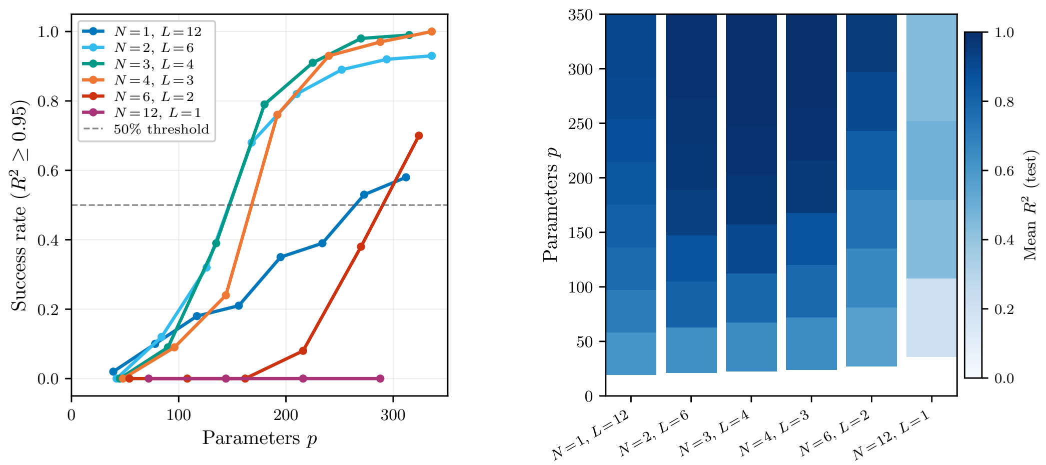

Figure 1: Trainability of data re-uploading QNNs at fixed encoding budget $E = NL = 12$, trained with Adam on 10 randomly drawn degree-12 Fourier targets. **Left:** success rate ($R^2 \geq 0.95$) vs. parameter count P. **Right:** mean R^2 (test) vs. P. All architectures share identical frequency spectra and redundancy structure, ruling out expressivity and spectral bias as explanations.

circuits are universal function approximators in the limit (Pérez-Salinas et al., 2021), expressivity alone cannot explain this failure. We identify **structural rank deficiency of the Jacobian** J as the mechanism responsible. For serial ($N = 1$) circuits, phase-locking forces all P gradient directions to share a common global phase, giving $\operatorname{rank}(J) \leq 2L + 1$ regardless of P — a property we term *structural gradient starvation* (distinct from gradient suppression in deep learning (Glorot & Bengio, 2010; Chen & He, 2021)): the fraction of parameters receiving identically zero gradient signal grows as $(P - \operatorname{rank}(J))/P \to 1$. Parallel architectures escape this via independent phase trajectories per qubit, maintaining $\sigma_{\min}(J^{(\mathrm{par})}) > 0$ generically for $P \leq 2E + 1$; the advantage is structural rather than threshold-based since both architectures share the same null-space onset $P > 2E + 1$ at fixed E. We make two contributions (Propositions 3.1–3.3, Corollary 3.3.1, Section 3): a proof of the rank ceiling and null-space growth for serial circuits, and a proof that parallel architectures avoid phase-locking via bilinear factorization. We further show that adding FM layers achieves $R^2 \geq 0.95$ with 1.6–2.2× fewer parameters than adding trainable blocks, *while* trainable blocks improve training only through the classical interpolation mechanism (Section 4).

2 Theoretical Background

2.1 Variational Quantum Circuits with Angle Encoding

Definition 2.1 (VQC with Angle Encoding)**.** An L-layer VQC with angle encoding consists of feature map (FM) operations $S(x) = e^{ixH}$, where $H = \frac{1}{2}\sigma$ for a Pauli operator $\sigma \in \{\sigma_x, \sigma_y, \sigma_z\}$, interleaved with parametrized ansatz layers $W_i(\boldsymbol{\theta}_i)$. The circuit unitary and scalar output are

$$U(x, \boldsymbol{\theta}) = W_L(\boldsymbol{\theta}_L) S(x) \cdots S(x) W_0(\boldsymbol{\theta}_0), \qquad f_{\boldsymbol{\theta}}(x) = \langle 0 | U^\dagger(x, \boldsymbol{\theta})\, M\, U(x, \boldsymbol{\theta}) | 0 \rangle, \tag{1}$$

where M is a fixed observable and $\boldsymbol{\theta} = (\boldsymbol{\theta}_0, \ldots, \boldsymbol{\theta}_L)$ are optimized by minimizing a training loss $\mathcal{L}(\boldsymbol{\theta})$. Note that an L-layer VQC contains $L + 1$ trainable ansatz blocks $W_0, \ldots, W_L$, sandwiching L encoding layers $S(x)$; parameter counts throughout this paper reflect this convention.

2.2 Fourier Representation, Spectral Invariance, and Spectral Bias

Theorem 2.1 (Fourier Representation (Schuld et al., 2021))**.** *The output of a VQC with L FM layers is a truncated Fourier series:*

$$f_{\boldsymbol{\theta}}(x) = \sum_{\omega \in \Omega} c_\omega(\boldsymbol{\theta})\, e^{i\omega x}, \tag{2}$$

with parameter-dependent coefficients $c_\omega(\boldsymbol{\theta}) \in \mathbb{C}$. Under unary encoding — *where each FM layer applies $S(x) = e^{ix\sigma/2}$ with unit scaling — the spectrum is $\Omega = \{-L, \ldots, L\}$ with $|\Omega| = 2L + 1$.*

Corollary 2.1.1 (Parameter Requirement (Schuld et al., 2021)). *Independent control of all Fourier coefficients requires $P \geq 2L+1$ ansatz parameters.*

Theorem 2.2 (Spectral Invariance (Holzer & Turkalj, 2024)). *For N qubits with L FM layers each, the spectrum depends only on the total encoding budget $E = NL$: serial ($N = 1$, $L = E$), parallel ($N = E$, $L = 1$), and hybrid architectures with entangling ansatz layers all generate identical frequency spectra $\Omega = \{-E, \dots, E\}$. Any trainability difference across architectures at fixed E is therefore a property of the ansatz structure, not of the function class.*

Under unary encoding, each frequency ω can be generated by $\binom{2E}{E-\omega}$ distinct FM combinations, a degeneracy that grows with E and introduces a spectral bias toward lower frequencies (Duffy & Jastrzebski, 2026), analogous to that in deep learning (Rahaman et al., 2019; Belis et al., 2026). This redundancy depends only on the total encoding budget E, and is therefore identical across architectures at fixed E.

2.3 Quantum Fisher Information Matrix

Definition 2.2 (Quantum Fisher Information Matrix (Stokes et al., 2020)). For $|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})|0\rangle$, the QFIM $\mathcal{F}(\boldsymbol{\theta}) \in \mathbb{R}^{P\times P}$ has entries

$$\mathcal{F}_{jk} = 4\,\mathrm{Re}[\langle \partial_j \psi | \partial_k \psi\rangle - \langle \partial_j \psi|\psi\rangle\langle\psi|\partial_k\psi\rangle]\,, \tag{3}$$

with rank equal to the number of independently explorable directions in state space at $\boldsymbol{\theta}$.

The maximum achievable QFIM rank is bounded above by the real dimension of the pure state manifold:

$$R_{\max} := \max_{\boldsymbol{\theta}} \mathrm{rank}(\mathcal{F}(\boldsymbol{\theta})) \leq 2^{N+1} - 2, \tag{4}$$

independently of P (Larocca et al., 2023). A VQC with $P \ll R_{\max}$ parameters operates in the underparameterized regime where the optimization landscape may contain spurious local minima; $P \geq R_{\max}$ is a necessary condition for landscape tractability in theory (Larocca et al., 2023), though empirically circuits often train successfully with fewer parameters. We invoke this bound in Appendix P to explain why the $N = 6$ minimum configuration ($P = 36$, $R_{\max} = 126$) starts well below saturation.

Our primary diagnostic — the Jacobian QFIM $\hat{\mathcal{F}} = \mathbf{J}_{\mathrm{data}}^{\top}\mathbf{J}_{\mathrm{data}}/n$ (Definition 2.3 and Section 2.4) — measures output sensitivity at discrete training points and has different rank bounds from $\mathcal{F}(\boldsymbol{\theta})$; it is the appropriate diagnostic for the function approximation setting studied here.

2.4 The Coefficient Matching Problem

By Theorem 2.1, training a VQC reduces to finding $\boldsymbol{\theta}^*$ such that $c_\omega(\boldsymbol{\theta}^*) = c^*_\omega$ for all $\omega \in \Omega$ — the *coefficient matching problem*. We assume the expressivity condition $\Omega_{\mathrm{target}} \subseteq \Omega_{\mathrm{model}}$ is satisfied throughout.

Definition 2.3 (Coefficient Matching Jacobian). Although $c_\omega(\boldsymbol{\theta}) \in \mathbb{C}$ (Theorem 2.1), the circuit output $f_{\boldsymbol{\theta}}(x) \in \mathbb{R}$ implies conjugate symmetry $c_{-\omega} = \overline{c_\omega}$, leaving $|\Omega| = 2L+1$ real degrees of freedom. We work with the *real representative* obtained by stacking independent coefficients:

$$\tilde{\mathbf{c}}(\boldsymbol{\theta}) = \left(\mathrm{Re}\,c_0,\ \mathrm{Re}\,c_1,\ \mathrm{Im}\,c_1,\ \dots,\ \mathrm{Re}\,c_L,\ \mathrm{Im}\,c_L\right)^{\top} \in \mathbb{R}^{2L+1}. \tag{5}$$

The *coefficient matching Jacobian* is then

$$J \in \mathbb{R}^{|\Omega|\times P}, \qquad J_{k,j} = \frac{\partial \tilde{c}_k(\boldsymbol{\theta})}{\partial \theta_j}, \tag{6}$$

with rows indexed by $k \in \{1, \dots, 2L+1\}$ corresponding to the real representatives above. Its rank governs the number of Fourier directions accessible to the optimizer: parameters in $\ker J$ are structurally decoupled from the loss and receive identically zero gradient signal.

The *data Jacobian* $\mathbf{J}_{\mathrm{data}} \in \mathbb{R}^{n\times P}$, with entries $(\mathbf{J}_{\mathrm{data}})_{i,j} = \partial f_{\boldsymbol{\theta}}(x_i)/\partial\theta_j$ evaluated at n training points $\{x_i\}_{i=1}^n$, measures sensitivity of the scalar circuit output over the training distribution. The *Jacobian QFIM* $\hat{\mathcal{F}} = \mathbf{J}_{\mathrm{data}}^{\top}\mathbf{J}_{\mathrm{data}}/n \in \mathbb{R}^{P\times P}$ is distinct from the coefficient Jacobian J: J governs the geometry of coefficient space and is the object studied in Sections 3–4, while $\hat{\mathcal{F}}$ measures output sensitivity at discrete training points and serves as the empirical diagnostic in Section 4. When $\operatorname{rank}(J) < P$, parameters in $\ker J$ produce no change in any Fourier coefficient and receive identically zero gradient signal regardless of initialization or training trajectory (Nocedal & Wright, 2006; Trefethen & Bau, 1997; Björck, 1996) — a structural phenomenon distinct from barren plateaus (McClean et al., 2018).

By Theorem 2.2, serial and parallel architectures at fixed E share the same $|\Omega|$ and hence the same dimensions of J; any difference in $\dim(\ker J)$ is purely structural, arising from how parameters couple to the Fourier coefficients — the subject of Section 3.

3 Structural Gradient Starvation in Serial Architectures

We now identify the structural property that distinguishes serial from parallel architectures at fixed encoding budget $E = NL$, characterizing it through a concrete comparison of the coefficient matching equations before proving the resulting rank and null-space bounds.

3.1 Structural Comparison: Serial vs. Parallel

We illustrate the structural difference at the smallest non-trivial encoding budget $E = 2$, where both architectures share spectrum $\Omega = \{-2, -1, 0, 1, 2\}$.

Serial ($N = 1$, $L = 2$). With three SU(2) parameter matrices $W^{(1)}, W^{(2)}, W^{(3)}$, writing $W^{(l)} = \left(\begin{smallmatrix} \alpha_l & -\bar{\beta}_l \\ \beta_l & \bar{\alpha}_l \end{smallmatrix}\right)$ for $l = 1, 2, 3$, the highest-frequency coefficient is:

$$c_{+2}^{\text{ser}} = -2\,\alpha_1\,\alpha_2^2\,\bar{\beta}_1\,\beta_3\bar{\alpha}_3. \tag{7}$$

All three matrices are coupled simultaneously; the interior matrix $W^{(2)}$ appears as α_2^2, re-entangling each gradient update with the current parameter value.

Parallel ($N = 2$, $L = 1$). With $\mathbf{v} = W^{(1)}|0\rangle \in \mathbb{C}^4$ and $\widetilde{M} = (W^{(2)})^\dagger M\, W^{(2)}$:

$$c_{+2}^{\text{par}} = v_1\bar{v}_4 \cdot \widetilde{M}_{4,1}. \tag{8}$$

Every summand is bilinear: the $W^{(1)}$-factor and the $W^{(2)}$-factor are completely separated. This pattern holds for all frequencies and generalizes to arbitrary N (Proposition A.1).

The general pattern. For general L and N, these structural differences persist: serial coefficients are degree-$2L + 2$ polynomials in the ansatz parameters with all matrices coupled, while parallel coefficients remain bilinear with $W^{(1)}$ and $W^{(2)}$ fully separated for all frequencies and all N (Proposition A.1, Appendix A; see Appendix L for a circuit diagram illustration).

Proposition 3.1 (Single-qubit gradient rank)**.** *For a single-qubit PQC, at any fixed $(x, \boldsymbol{\theta})$, $\nabla_{\boldsymbol{\theta}} f(x; \boldsymbol{\theta}) \in \mathbb{R}^P$ lies in a subspace of dimension at most 2, and $\operatorname{rank}(\mathcal{F}(\boldsymbol{\theta})) \leq 2$. Both bounds are tight for generic M and $|\psi\rangle$.*

Proof sketch. For $N = 1$, $|\psi\rangle$ lives on $\mathbb{CP}^1 \cong S^2$, real dimension 2. Its tangent space is spanned by a single complex vector $|\psi^\perp\rangle$, so $|\partial_j\psi\rangle = g_j|\psi^\perp\rangle + \lambda_j|\psi\rangle$. The gradient component reduces to $2\operatorname{Re}[\bar{g}_j\mu]$ where $\mu := \langle\psi^\perp|M|\psi\rangle \in \mathbb{C}$ is *shared across all* j. Hence $\nabla_{\boldsymbol{\theta}} f \in \operatorname{span}\{\operatorname{Re}(g), \operatorname{Im}(g)\}$, dimension ≤ 2. Tightness holds outside a measure-zero set of parameters, as shown in Appendix B. □

3.2 Serial Phase-Locking

Proposition 3.1 gives a 2D bound at each fixed x. It is *a priori* possible that the subspace $\mathcal{V}(x, \boldsymbol{\theta}) = \operatorname{span}\{\operatorname{Re}(g(x)), \operatorname{Im}(g(x))\}$ rotates as x varies and spans a larger subspace in J. The serial product structure prevents this.

Lemma 3.2 (Serial phase-locking)**.** *For a serial $N = 1$ re-uploading circuit:*

$$g_j(x) = h_j(x) \cdot G(x, \boldsymbol{\theta}), \tag{9}$$

where $G(x, \boldsymbol{\theta}) \in \mathbb{C}$ is a global phase factor common to all parameters and $h_j(x) \in \mathbb{C}$ depends only on the sub-circuit below layer j. The subspace $\mathcal{V}(x, \boldsymbol{\theta})$ rotates by the same phase for every parameter *as x varies.*

Proof sketch. For each layer j, $D_j^\dagger|\psi^\perp\rangle = e^{i\varphi}|\hat{\phi}_j^\perp\rangle$ where the phase φ is independent of j. We define $|\psi^\perp\rangle := T|\psi\rangle$ where $T = i\sigma_y K$ is the anti-unitary time-reversal operator, giving a canonically defined orthogonal complement that depends only on $|\psi\rangle$ and is shared across all j. With this gauge, φ depends only on the global state $|\psi\rangle$, establishing the phase independence. This yields $g_j = h_j \cdot e^{i\varphi}$, with h_j a trigonometric polynomial of degree $\leq L$ determined by the sub-circuit up to and including layer j. Full proof in Appendix C. ☐

Figure 2 provides direct empirical confirmation: gradient trajectories $g_j(x)$ from different ansatz blocks trace curves of identical shape in the complex plane for serial circuits ($\mathrm{Im}(g_j/g_k) = 0.000$ exactly), while parameters on different qubits in parallel circuits trace genuinely distinct trajectories ($|\mathrm{Im}(g_j/g_k)| = 0.19$ on average), as established in Corollary 3.3.1 below.

3.3 Structural Gradient Starvation in Serial $N = 1$ Circuits

Proposition 3.3 (Structural gradient starvation in serial circuits)**.** *Fix L and let P grow. For a serial single-qubit architecture:*

(i) $\mathrm{rank}(J) = 2L + 1$ *generically, so the rank ceiling imposed by the* $2L + 1$ *rows of J is tight and cannot be improved by adding parameters.*

(ii) For $P > 2L + 1$, $\dim(\ker J) \geq P - (2L + 1) \to \infty$ *as $P \to \infty$ at fixed L: the fraction of parameters receiving zero gradient signal,* $(P - \mathrm{rank}(J))/P \to 1$*, grows monotonically with P.*

Proof sketch. The bound $\mathrm{rank}(J) \leq 2L + 1$ follows from matrix dimensions since $J \in \mathbb{R}^{(2L+1)\times P}$. Tightness — $\mathrm{rank}(J) = 2L + 1$ generically — follows from the phase-locking of Lemma 3.2: each $h_j(x)$ is a trigonometric polynomial of degree $\leq L$, so all $\{h_j\}$ lie in $\mathcal{T}_L = \mathrm{span}\{e^{ikx} : |k| \leq L\}$, real dimension $2L + 1$; for generic $\boldsymbol{\theta}$ the $2L + 1$ rows of J are linearly independent, confirming that the ceiling is achieved. For $P > 2L + 1$, rank-nullity gives $\dim(\ker J) \geq P - (2L + 1) > 0$, growing without bound as $P \to \infty$ at fixed L. Every parameter in $\ker J$ produces no change in any Fourier coefficient and receives identically zero gradient signal, so the fraction of structurally decoupled parameters $(P - \mathrm{rank}(J))/P \to 1$, giving (ii). Full proof in Appendix D. ☐

Proposition 3.3 identifies $P = 2L + 1$ as the *Jacobian-rank optimum* for serial circuits: below this threshold the circuit cannot span all target Fourier directions; above it, null space growth dominates and any further improvement must come from the classical interpolation mechanism (Section 4) rather than expanded Fourier coverage. This structural gradient starvation arises from the phase-locking of Lemma 3.2 regardless of how parameters are added.

Corollary 3.3.1 (Parallel architecture: structural advantage via independent phase trajectories)**.** *Consider a parallel N-qubit architecture with L FM layers per qubit and encoding budget $E = NL$, with entangling ansatz layers and observable $M = Z_{N-1}$ as used in our experiments.*

(i) ***Onset threshold.*** *At fixed encoding budget $E = NL$, both serial and parallel architectures develop null directions in J when $P > 2E + 1$ (Proposition 3.3), consistopent with the shared frequency spectrum of Theorem 2.2. The trainability difference between architectures at fixed E therefore cannot be explained by a difference in onset threshold alone.*

(ii) ***Absence of phase-locking.*** *By Lemma 3.2, serial gradient starvation arises because all P parameters share a common global phase $G(x, \boldsymbol{\theta}) = e^{i\varphi}$, confining* $\mathrm{rank}(J^{(\mathrm{ser})}) \leq 2L + 1$ *independently of P. For $N > 1$ qubits, each qubit's local reduced state evolves with an independent local phase $\varphi^{(n)}$, since the downstream operator $D_j^{(n)}$ and state $|\psi^{(n)}\rangle$ differ*

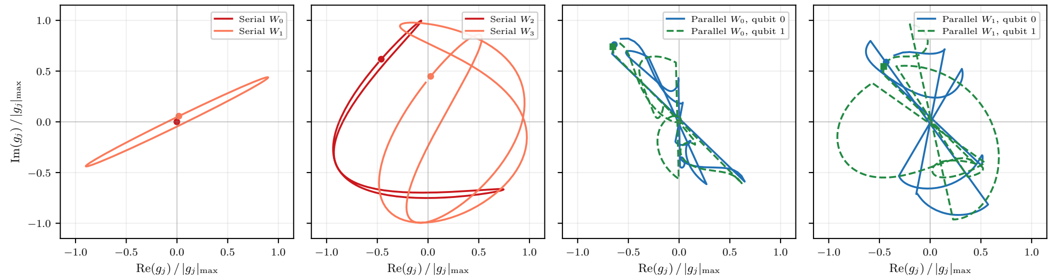

Figure 2: **Phase-locking in serial circuits vs. independent phase trajectories in parallel circuits.** Each curve shows $g_j(x) = \langle \psi^{\perp} | \partial_j \psi \rangle \in \mathbb{C}$, normalized by its maximum magnitude, as x sweeps $[0, 2\pi]$ and plotted in the complex plane. **Left two panels (serial, $N = 1$, $L = 4$):** parameters from the first four ansatz blocks $W_0, \ldots, W_3$ trace curves lying entirely on the real axis of $\mathbb{C}$, confirming that $g_j = h_j \in \mathbb{R}$ exactly (Lemma 3.2, Appendix C): the global phase factor $G = e^{i\varphi} = 1$ identically, so $g_j/g_k \in \mathbb{R}$ at every x ($\mathrm{Im}(g_j/g_k) = 0.000$ exactly). Curves are related by real scaling factors h_j/h_k, the direct signature of phase-locking. **Right two panels (parallel, $N = 2$, $L = 2$):** parameters on qubit 0 (solid) and qubit 1 (dashed) from the same ansatz block trace curves with genuinely different shapes that leave the real axis, reflecting independent local phases $\varphi^{(0)} \neq \varphi^{(1)}$ ($|\mathrm{Im}(g_j/g_k)| = 0.19$ on average). This holds consistently for both blocks W_0 and W_1, confirming it is a structural property of the parallel architecture rather than a parameter-specific coincidence.

across qubits. Parameters on different qubits are therefore modulated by independent phases, breaking the global coherence that produces the serial rank ceiling.

(iii) ***Full column rank below the threshold.*** *Because phase-locking is absent, the column space of $J^{(\mathrm{par})}$ is not confined to a single $\mathcal{T}_L$-subspace. For $P \leq 2E + 1$, $\sigma_{\min}(J^{(\mathrm{par})}) > 0$ generically, so all parameters contribute to at least one Fourier direction and no parameter lies in* $\ker J$*. By contrast, for serial circuits* $\dim(\ker J^{(\mathrm{ser})})/P \to 1$ *as $P \to \infty$, so an ever-growing fraction of parameters are structurally decoupled from the loss across the same range.*

We validate the rank ceiling empirically in Appendix K (Figure 5); a complete characterization of $\mathrm{rank}(J^{(\mathrm{par})})$ *for entangling ansätze remains an open problem.*

Remark 3.1 (Entangling ansätze)**.** The parallel advantage of Corollary 3.3.1 extends to the Rot-CNOT ladder used in our experiments: CNOT gates are unparameterized, so inter-qubit coupling does not grow with P, and phase-locking depends on the $\mathbb{CP}^1$ geometry of a single qubit and is broken by independent local phases regardless of entanglement. A full mechanistic justification is given in Appendix I.

Proof sketch. For $N > 1$ qubits, each qubit's local reduced state evolves with an independent local phase $\varphi^{(n)}$, since the downstream operator $D_j^{(n)}$ and state $|\psi^{(n)}\rangle$ differ across qubits. The global coherence condition of Lemma 3.2 is therefore broken: columns of $J^{(\mathrm{par})}$ are modulated by independent phases and are not confined to a single $\mathcal{T}_L$-subspace, giving $\sigma_{\min}(J^{(\mathrm{par})}) > 0$ generically for $P \leq 2E+1$ by a real-analyticity argument. Null directions appear at $P > 2E+1$ by rank-nullity applied to the shared spectrum $|\Omega| = 2E + 1$, the same threshold as the serial case at fixed E. Full proof in Appendix E. $\square$

Summary. Phase-locking in serial circuits confines all P gradient directions to $\mathcal{T}_L$, as confirmed directly by Figure 2. Together, Proposition 3.3 and Corollary 3.3.1 establish the key separation (Theorem F.1, Appendix F): serial gradient starvation sets in at $P > 2L + 1$ regardless of parameter count, while parallel architectures maintain $\sigma_{\min}(J^{(\mathrm{par})}) > 0$ generically for $P \leq 2E + 1$, deferring null space growth to the same threshold $P > 2E + 1$ that applies to serial circuits at fixed E.

4 FM Layers vs. Trainable Blocks: A Structural Comparison

Section 3 establishes the rank ceiling $2NL+1$ as the fundamental structural constraint on trainability: serial circuits are confined to this ceiling by phase-locking, while parallel circuits maintain full column rank up to it. This ceiling is not fixed — it is a property of the encoding structure, not the ansatz, and can be raised by adding FM layers. This observation directly motivates the central question of this section: given a parallel circuit where gradient starvation is not yet the bottleneck, which route to more parameters is more effective — adding *feature map (FM) layers*, which increase P while simultaneously expanding the Fourier spectrum and raising the rank ceiling, or adding *trainable blocks*, which increase P without changing either? This section shows these two routes have fundamentally different effects on trainability, with the FM route achieving $R^2 \geq 0.95$ with 1.6–$2.2\times$ fewer parameters across all tested architectures. Full experimental details and degree justification are given in Appendices I and P.

4.1 Two Routes to More Parameters

Starting from a circuit with L FM layers and one trainable block layer per encoding block, there are two natural ways to add parameters while keeping the architecture parallel:

1. **FM route**: increase the number of FM layers L, keeping the trainable block depth $\mathrm{tbl} = 1$ fixed. Each extra FM layer adds $N \cdot 3$ parameters and simultaneously expands $|\Omega|$, raising the Jacobian rank ceiling and making more Fourier directions accessible to the optimizer.
2. **Trainable blocks route**: increase the number of trainable block layers tbl, keeping $L = L_{\min}$ fixed. Each extra block adds $(L_{\min} + 1) \cdot N \cdot 3$ parameters without changing the Fourier spectrum or redundancy structure. The Jacobian rank ceiling remains fixed.

As established in Corollary 3.3.1, parallel circuits have a rank ceiling of $|\Omega| = 2E + 1$ beyond which additional parameters create null directions in J. Adding FM layers raises this ceiling; adding trainable blocks increases P without raising the ceiling, relying on the classical interpolation mechanism ($P \geq n_{\mathrm{train}}$) with no quantum-specific benefit. Both routes assume $L \geq L_{\min} = \lceil \mathrm{deg}/N \rceil$ FM layers are already present; below $L_{\min}$ the target frequencies lie outside the circuit's function class and neither route can improve training. In all experiments we fix $L = L_{\min}$ for the trainable blocks route and $\mathrm{tbl} = 1$ for the FM route, isolating the effect of each.

4.2 The FM Route: Rank Growth

Figure 3 shows both routes for $N \in \{1, 6\}$, representing contrasting efficiency regimes: $N = 1$ (strong gradient starvation, large target degree relative to encoding budget) and $N = 6$ (mild gradient starvation, target degree well within the circuit's accessible spectrum at $L_{\min}$). As L increases from $L_{\min}$, the normalized Jacobian QFIM eigenvalue spectra λ_i/λ_1 shift visibly to the right. We characterize this by the *spectral knee* — the rank index beyond which λ_i/λ_1 drops sharply — a threshold-free measure of the number of genuinely exploitable gradient directions. Along the FM route, the knee generally shifts rightward with each extra layer: at $L = 12$ the spectrum drops sharply at rank 25; by $L = 22$ this extends to rank 33. Along the trainable blocks route, the spectral knee is frozen at the theoretical rank ceiling $2NL_{\min} + 1$ for all tested architectures: $N = 1$ freezes at rank 24–25 (ceiling 25), $N = 2$ at rank ≈ 41 (ceiling 41), $N = 4$ at rank ≈ 57 (ceiling 57), and $N = 6$ at rank 12–13 (ceiling 13) (Appendix O, Figure 3 bottom right). This confirms that adding trainable blocks provides access to no new Fourier directions regardless of architecture: the rank ceiling imposed by $L_{\min}$ is tight, and the tbl route can only improve coverage of directions already within that ceiling — a purely classical interpolation mechanism with no quantum-specific benefit.

Spectral knee as a trainability diagnostic. The spectral knee — the rank index beyond which λ_i/λ_1 drops sharply — distinguishes the two routes structurally. Along the FM route, the knee grows monotonically with each extra layer because the ceiling $2NL+1$ itself expands: each FM layer makes genuinely new Fourier directions accessible to the optimizer. Along the trainable blocks route, the ceiling is fixed at $2NL_{\min} + 1$; the knee is frozen at this ceiling throughout the tbl sweep, confirming that no new Fourier directions become accessible via the classical interpolation mechanism. This distinction is quantified in Appendix N (Figure 9).

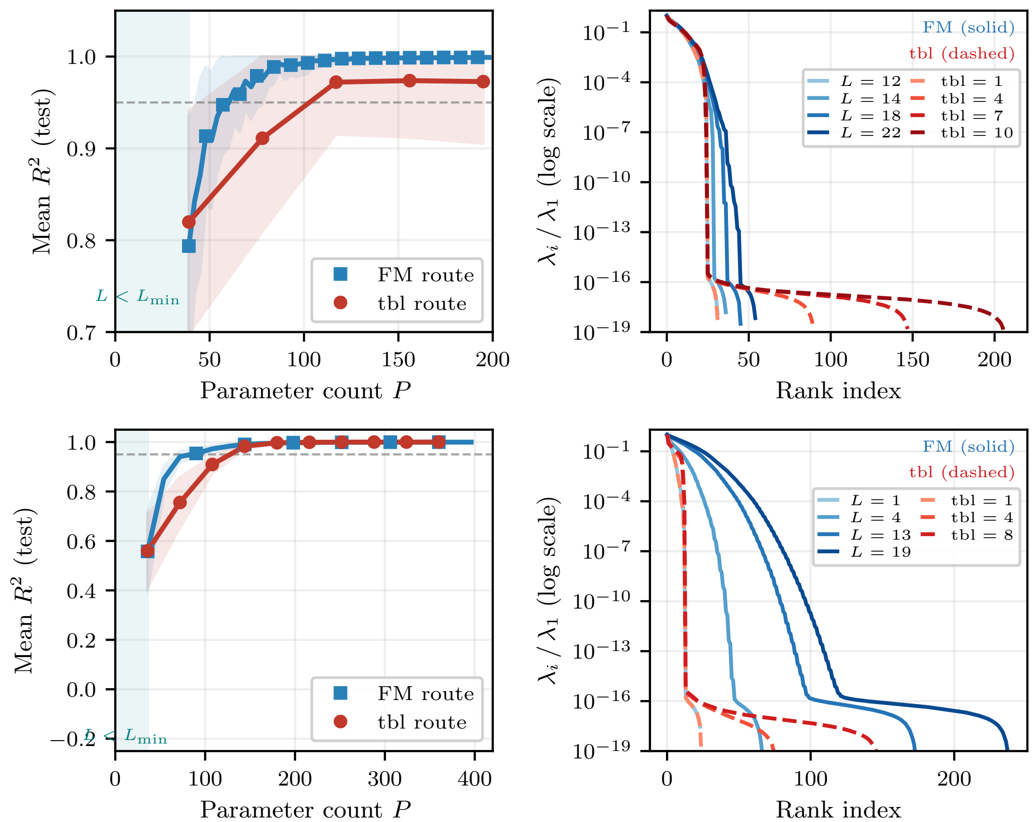

Figure 3: **FM route vs. trainable blocks route for** $N = 1$ **(gradient starvation) and** $N = 6$ **(high-qubit regime). Left panels:** mean R^2 (test) vs. parameter count P for the FM route (blue, solid) and trainable blocks route (red, dashed). Teal shading: expressivity gap ($L < L_{\min}$). Dashed gray line: $R^2 = 0.95$ threshold. **Right panels:** normalized Jacobian QFIM eigenvalue spectra λ_i/λ_1 at selected L values (FM, solid blue) and tbl values (trainable blocks, dashed red). *Top row* ($N = 1$)*:* the spectral knee shifts rightward with each extra FM layer while remaining frozen at the theoretical rank ceiling $2L_{\min} + 1 = 25$, despite R^2 improving along both routes. *Bottom row* ($N = 6$)*:* both routes reach $R^2 \geq 0.95$. The spectral knee shifts rightward along the FM route and remains frozen along the tbl route, consistent with the structural prediction of Section 4.2. Parameter efficiency ratios for all architectures are given in Table 2.

4.3 The Trainable Blocks Route: Interpolation Threshold

Despite the spectral knee being frozen at the fixed $L_{\min}$ ceiling,, R^2 still improves substantially with tbl. The governing mechanism is the classical interpolation threshold: once the parameter count P exceeds the training set size n_{train}, the system $f(x_i; \boldsymbol{\theta}) = y_i$ becomes overdetermined and gradient descent finds an interpolating solution (Belkin et al., 2019; Bartlett et al., 2020). This is a purely classical phenomenon with no dependence on quantum structure. The spectral knee of $\hat{\mathcal{F}}$ remains frozen at the $L_{\min}$ ceiling throughout the tbl sweep, confirming that no new Fourier directions become accessible as trainable blocks are added. For $N = 6$, deg$= 6$, both routes reach $R^2 \geq 0.95$: the FM route at $L^* = 4$ ($P = 90$) and the trainable blocks route at $\mathrm{tbl}^* = 4$ ($P = 144$), giving a $1.6\times$ efficiency ratio. The lower ratio relative to $N = 1, 2, 4$ reflects that the circuit is already expressive enough at $L_{\min} = 1$ for a degree-6 target, so extra FM layers provide less marginal benefit than at higher target degrees.

Table 1 (Appendix G) shows that $R^2 \geq 0.95$ is first achieved within 1–3 tbl steps of the interpolation threshold $P \geq n_{\mathrm{train}} = 200$ across all architectures that reach saturation, consistent with this mechanism.

4.4 Efficiency Comparison

Table 2 (Appendix H) quantifies the parameter efficiency of both routes for architectures where both reach saturation. The FM route requires 1.6–$2.2\times$ fewer parameters across all four architectures: each new FM layer raises the rank ceiling, keeping a growing fraction of parameters exploitable, while the tbl route keeps the ceiling fixed and relies on classical interpolation. This advantage spans $N = 1$ to $N = 6$ and target degrees from 6 to 28, confirming it reflects a structural property of the FM route rather than an architecture-dependent accident. The FM route is the only route that engages a quantum-specific mechanism: it simultaneously expands the accessible frequency spectrum and shifts the spectral knee rightward in a way that has no classical analog.

4.5 Validation on Real-World Data

The structural predictions of Sections 3 and 4 are validated on the Nottingham temperature dataset — 240 monthly mean air temperatures recorded at Nottingham Castle from 1920 to 1939 — in Appendix M, using two experiments that probe different regimes.

With encoding budget $E = N \times L_{\min} = 12$ (Figure 7), the model spectrum is insufficient at $L_{\min}$ for all architectures: the Nottingham spectrum extends beyond $L_{\min}$ coverage, so FM layers provide a dual benefit — expanding Jacobian rank coverage *and* expressivity — while the trainable blocks route fails categorically ($R^2 < 0$ throughout) for all four architectures, unable to expand either.

With encoding budget $E = N \times L_{\min} = 24$ (Figure 8), expressivity is sufficient at $L_{\min}$ and the experiment isolates parameter efficiency. The FM route achieves 1.9–$2.7\times$ fewer parameters than the tbl route for $N \in \{1, 2, 4\}$, consistent with the synthetic results of Table 2. For $N = 6$, the result inverts: the tbl route reaches $R^2 \geq 0.95$ while the FM route plateaus at $R^2 \approx 0.83$, consistent with the $N = 6$ circuit entering the Larocca underparameterization regime $P \approx R_{\max} = 126$ (Larocca et al., 2023) early in the FM sweep, where adding encoding layers increases the parameter demand faster than the added parameters satisfy it.

5 Conclusion

The trainability gap between serial and parallel QNN architectures at fixed encoding budget $E = NL$ has a precise mechanistic explanation. The $\mathbb{CP}^1$ geometry of a single-qubit state space forces all P gradient directions to share a common global phase (Lemma 3.2, Figure 2), confining the column space of the coefficient matching Jacobian J to a subspace of dimension $2L + 1$: as P grows at fixed L, the fraction of parameters receiving identically zero gradient signal approaches 1 — *structural gradient starvation*. Parallel architectures defer this via independent phase trajectories across qubits, maintaining $\sigma_{\min}(J^{(\mathrm{par})}) > 0$ generically for $P \leq 2E + 1$, so every parameter contributes to at least one exploitable Fourier direction up to the shared onset threshold.

Add FM layers, not trainable blocks. The structural analysis directly motivates the practical recommendation. FM layers simultaneously expand the accessible Fourier spectrum and shift the spectral knee of the Jacobian QFIM rightward — a quantum-specific mechanism with no classical analog. Trainable blocks increase P without changing either the spectrum or the spectral knee, improving training solely via the classical interpolation threshold ($P \geq n_{\mathrm{train}}$). The 1.6–$2.2\times$ parameter efficiency advantage of the FM route is a direct consequence of this structural difference: FM parameters are added below the rank ceiling, while trainable block parameters accumulate in $\ker J$.

Limitations and future work. Theoretical results are proved for the architecture extremes ($N = 1$ serial, product ansatz parallel); extension to hybrid architectures and general entangling ansätze remains open. The efficiency advantage of the FM route is characterized for circuits operating below the geometric threshold $R_{\max} = 2^{N+1} - 2$ of Larocca et al. (2023); in this underparameterized regime, trainable blocks may offer competitive parameter efficiency by better exploiting existing state-space directions before the accessible spectrum is fully saturated, and a more complete characterization of the FM vs. trainable blocks trade-off in this regime remains open. Real-world validation on the Nottingham temperature dataset confirms the structural predictions and identifies the boundary of the FM efficiency advantage. With encoding budget $E = N \times L_{\min} = 12$, the trainable blocks route fails categorically for all architectures while the FM route succeeds by simultaneously expanding expressivity and Jacobian rank coverage. With $E = N \times L_{\min} = 24$, the FM route achieves 1.9–2.7$\times$ fewer parameters than the tbl route for $N \in \{1, 2, 4\}$, consistent with the synthetic results. The exception is $N = 6$, where the FM advantage inverts: the tbl route reaches $R^2 \geq 0.95$ while the FM route plateaus, consistent with the circuit entering the Larocca underparameterization regime early in the FM sweep. This identifies a natural boundary condition for the FM efficiency advantage and motivates a more complete characterization of the FM vs. trainable blocks trade-off in the Larocca-limited regime as future work.

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A Proof of Proposition A.1

Proposition A.1 (Bilinear factorization for general parallel architectures)**.** *For a parallel N-qubit architecture with $L = 1$ encoding layer, the Fourier coefficients of the circuit output are:*

$$c_{\omega}^{(\mathrm{par})} = \sum_{\substack{j,k \in \{0,\dots,2^N-1\} \\ \sum_n (k_n - j_n) = \omega}} \bar{v}_k \, \widetilde{M}_{kj} \, v_j, \tag{10}$$

where $\mathbf{v} = W^{(1)}|0\rangle^{\otimes N}$ and $\widetilde{M} = (W^{(2)})^{\dagger} M \, W^{(2)}$. Every term is bilinear*: $\bar{v}_k v_j$ depends only on $W^{(1)}$ and $\widetilde{M}_{kj}$ only on $W^{(2)}$, with no cross-coupling between the two ansatz layers.*

Proof. **Step 1: Diagonal encoding and circuit output.** Since any Pauli operator σ is unitarily equivalent to σ_z, the FM layer $S(x) = e^{ixH}$ with $H = \frac{1}{2}\sigma$ is diagonal in an appropriate single-qubit basis, with eigenvalues $\pm\frac{1}{2}$. Working in this basis, $S(x)|k\rangle = e^{i\phi_k(x)}|k\rangle$ where

$$\phi_k(x) = \frac{x}{2} \sum_{n=1}^{N} (1 - 2k_n), \tag{11}$$

with $k_n \in \{0, 1\}$ the n-th bit of k. Setting $\mathbf{v} = W^{(1)}|0\rangle^{\otimes N}$ and $\widetilde{M} = (W^{(2)})^{\dagger} M \, W^{(2)}$, the circuit output expands as:

$$f(x; \boldsymbol{\theta}) = \langle \mathbf{v} | \, e^{-ixH} \widetilde{M} \, e^{ixH} | \mathbf{v} \rangle = \sum_{j,k=0}^{2^N-1} \bar{v}_k \, \widetilde{M}_{kj} \, v_j \, e^{i(\phi_k(x) - \phi_j(x))}. \tag{12}$$

Step 2: Frequency decomposition. The Fourier frequency of term (j, k) is:

$$\omega_{jk} = \phi_k(x)/x - \phi_j(x)/x = \sum_{n=1}^{N} (k_n - j_n) \in \{-N, \dots, N\}. \tag{13}$$

Collecting all terms with the same frequency ω gives:

$$c_{\omega} = \sum_{\substack{j,k \in \{0,\dots,2^N-1\} \\ \sum_n (k_n - j_n) = \omega}} \bar{v}_k \, \widetilde{M}_{kj} \, v_j. \tag{14}$$

This sum includes all pairs (j, k) with the appropriate net qubit difference, including non-complementary pairs where $j_n = k_n$ at some positions.

Step 3: Bilinearity. Every term in (14) factorizes as:

$$\bar{v}_k \, \widetilde{M}_{kj} \, v_j = \underbrace{\bar{v}_k \, v_j}_{W^{(1)}\text{-factor}} \cdot \underbrace{\widetilde{M}_{kj}}_{W^{(2)}\text{-factor}}. \tag{15}$$

The entries v_j, v_k of $\mathbf{v} = W^{(1)}|0\rangle^{\otimes N}$ depend only on $W^{(1)}$ and are independent of $W^{(2)}$. The entries $\widetilde{M}_{kj}$ of $(W^{(2)})^{\dagger} M \, W^{(2)}$ depend only on $W^{(2)}$ and are independent of $W^{(1)}$. No term couples $W^{(1)}$ and $W^{(2)}$ multiplicatively, establishing the bilinear separation for all frequencies $\omega \in \{-N, \dots, N\}$. The $N = 2$ case recovers equation (8) directly. □

B Full Proof of Proposition 3.1

Proof. We work at fixed $(x, \boldsymbol{\theta})$, writing $|\psi\rangle = |\psi(x, \boldsymbol{\theta})\rangle$.

Step 1: Tangent space of $\mathbb{CP}^1$. For $N = 1$, normalized pure states form $\mathbb{CP}^1 \cong S^2$, a smooth real-2-dimensional manifold (Bengtsson & Życzkowski, 2006). At any $|\psi\rangle$, the tangent space $T_{|\psi\rangle}\mathbb{CP}^1$ is spanned over $\mathbb{C}$ by the unique (up to phase) unit vector $|\psi^{\perp}\rangle$ with $\langle \psi | \psi^{\perp} \rangle = 0$. Any infinitesimal variation satisfies:

$$|\partial_j \psi\rangle = g_j |\psi^{\perp}\rangle + \lambda_j |\psi\rangle, \qquad g_j \in \mathbb{C}, \ \lambda_j \in i\mathbb{R}, \tag{16}$$

where $\lambda_j \in i\mathbb{R}$ follows from $\langle\psi|\psi\rangle = 1$.

Step 2: The gradient of f. $f = \langle\psi|M|\psi\rangle$ for Hermitian M. Differentiating and using $\bar{\lambda}_j\langle\psi|M|\psi\rangle \in i\mathbb{R}$ (vanishes under $\mathrm{Re}[\cdot]$):

$$\frac{\partial f}{\partial\theta_j} = 2\,\mathrm{Re}[\bar{g}_j\langle\psi^\perp|M|\psi\rangle]. \tag{17}$$

Step 3: Shared scalar μ. Define $\mu := \langle\psi^\perp|M|\psi\rangle \in \mathbb{C}$, which is the same for every j. Writing $\mu = a + ib$ and $g_j = u_j + iw_j$:

$$\frac{\partial f}{\partial\theta_j} = 2(au_j + bw_j) = 2\big[a\,\mathrm{Re}(g_j) + b\,\mathrm{Im}(g_j)\big]. \tag{18}$$

Hence $\nabla_{\boldsymbol{\theta}} f = 2a\,\mathrm{Re}(g) + 2b\,\mathrm{Im}(g) \in \mathrm{span}\{\mathrm{Re}(g), \mathrm{Im}(g)\}$, which has dimension ≤ 2.

Step 4: Rank of the QFIM.

$$\mathcal{F}_{jk} = 4\,\mathrm{Re}[\bar{g}_j g_k] = 4\big(\mathrm{Re}(g)_j\mathrm{Re}(g)_k + \mathrm{Im}(g)_j\mathrm{Im}(g)_k\big), \tag{19}$$

a sum of two rank-one matrices (Stokes et al., 2020), so $\mathrm{rank}(\mathcal{F}) \leq 2$.

Tightness. For generic M and $|\psi\rangle$, $\mu \neq 0$ and $g \notin \mathbb{R}^P$, so $\mathrm{Re}(g)$ and $\mathrm{Im}(g)$ are linearly independent and the bound is tight. ☐

C Full Proof of Lemma 3.2

Proof. Fix $(x, \boldsymbol{\theta})$. For each $j \in \{1, \ldots, L\}$ define:

$$D_j := \prod_{l=L}^{j+1} V_l(\boldsymbol{\theta}_l) S_l(x) \in \mathrm{SU}(2), \tag{20}$$

$$|\phi_j\rangle := V_j(\boldsymbol{\theta}_j)\, S_j(x) \prod_{l=j-1}^{1} V_l(\boldsymbol{\theta}_l)\, S_l(x)|\psi_0\rangle \in \mathbb{C}^2, \tag{21}$$

so that $|\psi\rangle = D_j|\phi_j\rangle$ and $|\partial_j\psi\rangle = D_j(\partial_j V_j)V_j^{-1}|\phi_j\rangle$.

Step 1: Gradient coefficient. Since $|\psi\rangle = D_j|\phi_j\rangle$ and $D_j \in \mathrm{SU}(2)$ is unitary, the orthogonal complement transforms as $|\psi^\perp\rangle = D_j|\hat{\phi}_j^\perp\rangle$, giving $D_j^\dagger|\psi^\perp\rangle = |\hat{\phi}_j^\perp\rangle$. Differentiating $|\psi\rangle = D_j|\phi_j\rangle$ with respect to θ_j and using the fact that D_j does not depend on θ_j (it is the sub-circuit *above* layer j):

$$|\partial_j\psi\rangle = D_j\,(\partial_j V_j)\,V_j^{-1}|\phi_j\rangle. \tag{22}$$

The gradient coefficient is therefore:

$$g_j = \langle\psi^\perp|\partial_j\psi\rangle = \langle\psi^\perp|\,D_j\,(\partial_j V_j)V_j^{-1}|\phi_j\rangle = \langle D_j^\dagger\psi^\perp|(\partial_j V_j)V_j^{-1}|\phi_j\rangle, \tag{23}$$

where the last equality uses unitarity of D_j. The key observation is that the right-hand factor $(\partial_j V_j)V_j^{-1}|\phi_j\rangle$ depends only on the sub-circuit *up to and including* layer j, while $D_j^\dagger|\psi^\perp\rangle$ depends only on the sub-circuit *above* layer j. Step 2 shows that the latter is independent of j up to a global phase.

Step 2: Canonical gauge and phase independence. To eliminate the $\mathrm{U}(1)$ ambiguity in the definition of $|\psi^\perp\rangle$, we fix a canonical gauge via the anti-unitary time-reversal operator $T = i\sigma_y K$, setting $|\psi^\perp\rangle := T|\psi\rangle$ and $|\hat{\phi}_j^\perp\rangle := T|\hat{\phi}_j\rangle$ for all j. This choice is $\mathrm{U}(1)$-equivariant: if $|\psi\rangle \mapsto e^{i\alpha}|\psi\rangle$ then $T|\psi\rangle \mapsto e^{-i\alpha}T|\psi\rangle$, so relative phases between $|\psi^\perp\rangle$ and $|\hat{\phi}_j^\perp\rangle$ are preserved independently of j. For any $D \in \mathrm{SU}(2)$ with $D = \left(\begin{smallmatrix} a & -\bar{b} \\ b & \bar{a} \end{smallmatrix}\right)$, direct computation gives

$$(i\sigma_y)\,D^* = D\,(i\sigma_y), \tag{24}$$

which implies $TD = DT$ and hence $TD_j^\dagger = D_j^\dagger T$. Using this together with $|\psi^\perp\rangle = T|\psi\rangle$:

$$D_j^\dagger|\psi^\perp\rangle = D_j^\dagger T|\psi\rangle = TD_j^\dagger|\psi\rangle = T|\hat{\phi}_j\rangle = |\hat{\phi}_j^\perp\rangle, \tag{25}$$

where the third equality uses $D_j^\dagger|\psi\rangle = |\hat{\phi}_j\rangle$ by construction. The phase factor is therefore:

$$e^{i\varphi_j} = \langle T\hat{\phi}_j \,|\, D_j^\dagger T\psi\rangle = \langle T\hat{\phi}_j \,|\, TD_j^\dagger\psi\rangle = \overline{\langle\hat{\phi}_j \,|\, D_j^\dagger\psi\rangle} = \overline{\langle\hat{\phi}_j \,|\, \hat{\phi}_j\rangle} = 1, \tag{26}$$

where the third equality uses the anti-unitarity of T: $\langle Ta \,|\, Tb\rangle = \overline{\langle a \,|\, b\rangle}$. Hence $G = e^{i\varphi} = 1$ independently of j, x, and $\boldsymbol{\theta}$: the global phase factor is identically unity and each $g_j = h_j$ is real-valued.

Step 3: Factorization. Since $D_j^\dagger|\psi^\perp\rangle = |\hat{\phi}_j^\perp\rangle$ (Step 2), substituting back:

$$g_j(x) = \underbrace{\langle\hat{\phi}_j^\perp|(\partial_j V_j)V_j^{-1}|\phi_j\rangle}_{h_j(x)\,\in\,\mathbb{R}}, \tag{27}$$

establishing $g_j = h_j$ with h_j real-valued. The per-input gradient rank bound ≤ 2 then follows from Proposition 3.1.

Step 4: Phase-locking and rank of J. Since $g_j = h_j \in \mathbb{R}$ for all j, the subspace $\mathcal{V}(x, \boldsymbol{\theta}) = \operatorname{span}\{\operatorname{Re}(g), \operatorname{Im}(g)\}$ collapses: $\operatorname{Im}(g_j) = 0$ identically, so $g_j/g_k \in \mathbb{R}$ at every x — the phase-locking of Lemma 3.2 is exact, not merely approximate. Each h_j is determined by j encoding gates $S_1, \ldots, S_j$ and is therefore a trigonometric polynomial of degree $\leq j \leq L$ (Schuld et al., 2021). All $\{h_j\}_{j=1}^P$ therefore lie in $\mathcal{T}_L = \operatorname{span}\{e^{ikx} : |k| \leq L\}$, a real vector space of dimension $\leq 2L+1$. Since φ is common to all columns, the column space of J has dimension $\leq 2L+1$ independently of P. $\square$

D Full Proof of Proposition 3.3

Proof. **Step 1: Jacobian entry.** From Proposition 3.1 and Lemma 3.2, substituting $g_j = h_j e^{i\varphi}$:

$$J_{\omega,j} = \int_0^{2\pi} \alpha(x)\operatorname{Re}(h_j)\, e^{-i\omega x}\, dx + \int_0^{2\pi} \beta(x)\operatorname{Im}(h_j)\, e^{-i\omega x}\, dx, \tag{28}$$

where $\alpha(x) := 2(a\cos\varphi + b\sin\varphi)$ and $\beta(x) := 2(b\cos\varphi - a\sin\varphi)$ are independent of j, with $a, b := \operatorname{Re}(\mu), \operatorname{Im}(\mu)$ from Proposition 3.1.

Step 2: Degree bound on h_j. Each encoding gate $S_l(x) = e^{ix\sigma/2}$ contributes one factor of $e^{\pm ix/2}$, so $h_j \in \mathcal{T}_L$ with $h_j(x) = \sum_{|k|\leq j} \hat{h}_j^{(k)} e^{ikx}$ and $j \leq L$ (Schuld et al., 2021). All P columns of J therefore lie in $\mathcal{T}_L$.

Step 3: Rank bound and tightness (i). The bound $\operatorname{rank}(J) \leq 2L+1$ follows from $J \in \mathbb{R}^{(2L+1)\times P}$. Tightness holds because the map $h_j \mapsto J_{\cdot,j}$ is a Fourier projection into $\mathcal{T}_L$, and for generic $\boldsymbol{\theta}$ the $2L+1$ rows of J corresponding to distinct frequencies $\omega \in \Omega$ are linearly independent — established by the real-analyticity argument applied to $\det(JJ^\top)$, which is not identically zero since the circuit is expressive (Schuld et al., 2021). Hence $\operatorname{rank}(J) = 2L+1$ generically, proving (i).

Step 4: Growth of the null space (ii). For $P > 2L+1$, the rank-nullity theorem applied to Step 3 gives:

$$\dim(\ker J) = P - \operatorname{rank}(J) \geq P - (2L+1), \tag{29}$$

which grows without bound as $P \to \infty$ at fixed L. Every parameter $\boldsymbol{\theta}_j \in \ker J$ satisfies $J_{\omega,j} = 0$ for all $\omega \in \Omega$, meaning it produces no change in any Fourier coefficient and receives identically zero gradient signal. The fraction of such parameters,

$$\frac{\dim(\ker J)}{P} \geq \frac{P - (2L+1)}{P} \to 1 \quad \text{as } P \to \infty, \tag{30}$$

grows monotonically toward 1, so the overwhelming majority of parameters become structurally decoupled from the loss as P increases at fixed L. This is independent of the condition number of the exploitable subspace $\operatorname{im}(J^\top)$, which has dimension at most $2L+1$ throughout. $\square$

Remark D.1 (Encoding-dependence vs. architecture-dependence)**.** Spectral bias is *encoding-dependent*: redundancy numbers $\binom{2E}{E-\omega}$ crowd out high-frequency gradients under unary encoding but are reduced under richer schemes (e.g. Chebyshev or non-uniform encodings). Structural gradient starvation is *architecture-dependent*: the $\mathbb{CP}^1$ geometry bounds per-input gradient rank at 2 regardless of encoding, and phase-locking caps the Jacobian rank at $2L_{\text{eff}}+1$ where L_{eff} scales with encoding richness. Richer encodings therefore shift *when* structural gradient starvation sets in, but not *whether* it does.

E Full Proof of Corollary 3.3.1

Proof. **Step 1: Structure of $J^{(\text{par})}$ and the phase-locking argument.** For a product ansatz $U = \bigotimes_n U^{(n)}$ with a tensor product observable, the global Fourier coefficients are convolutions of local coefficients:

$$c_\omega = \sum_{\omega_1+\cdots+\omega_N=\omega} \prod_{n=1}^{N} c^{(n)}_{\omega_n}. \tag{31}$$

Differentiating with respect to $\theta^{(m)}_j$ yields

$$\frac{\partial c_\omega}{\partial \theta^{(m)}_j} = \sum_{\omega_1+\cdots+\omega_N=\omega} \frac{\partial c^{(m)}_{\omega_m}}{\partial \theta^{(m)}_j} \cdot \prod_{n\neq m} c^{(n)}_{\omega_n}, \tag{32}$$

which depends on the coefficients of all qubits $n \neq m$ via $\prod_{n\neq m} c^{(n)}_{\omega_n}$, so $J^{(\text{par})}$ is generically dense.

The parallel advantage is instead established via the phase-locking argument of Lemma 3.2. For $N = 1$, the phase $\varphi(x, \boldsymbol{\theta})$ in $g_j = h_j e^{i\varphi}$ is determined by the unique state $|\psi\rangle \in \mathbb{CP}^1$ and is identical for all P parameters, confining $\operatorname{rank}(J^{(\text{ser})}) \leq 2L + 1$ regardless of P. For $N > 1$, each qubit n has its own downstream operator $D^{(n)}_j$ and local reduced state $|\psi^{(n)}\rangle$, so the local phase $\varphi^{(n)}$ is qubit-dependent. Parameters on qubit m are modulated by $e^{i\varphi^{(m)}}$ while parameters on qubit $n \neq m$ are modulated by $e^{i\varphi^{(n)}}$, with $\varphi^{(m)} \neq \varphi^{(n)}$ generically. The global coherence condition of Lemma 3.2 is therefore broken: columns of $J^{(\text{par})}$ corresponding to different qubits are modulated by independent phases and are not confined to a single $\mathcal{T}_L$-subspace.

Step 2: Onset threshold at fixed E. The shared frequency spectrum $\Omega = \{-E, \dots, E\}$ (Theorem 2.2) implies $|\Omega| = 2E + 1$ rows in $J^{(\text{par})}$ regardless of N. By rank-nullity:

$$\dim(\ker J^{(\text{par})}) \geq P - (2E + 1) > 0 \quad \text{for } P > 2E + 1, \tag{33}$$

so null directions appear at threshold $P \approx 2E + 1$, consistent with Proposition 3.3.

Step 3: Full column rank for $P \leq 2E + 1$. Since the column space of $J^{(\text{par})}$ spans more than $2L+1$ dimensions generically — by absence of phase-locking — $\sigma_{\min}(J^{(\text{par})}) > 0$ for $P \leq 2E+1$. Formally, the map $\Phi : \mathbb{R}^P \to \mathbb{R}^{2E+1}$ from parameters to real Fourier representatives (Definition 2.3) is real-analytic, and the circuit is expressive for $P \leq 2E + 1$, so $q(\boldsymbol{\theta}) := \det(J^{(\text{par})\top} J^{(\text{par})})$ is not identically zero. By the real-analyticity argument (Bochnak et al., 1998, Prop. 2.2.7), $q > 0$ for almost all $\boldsymbol{\theta}$, giving $\sigma_{\min}(J^{(\text{par})}) > 0$ generically, so no parameter lies in $\ker J^{(\text{par})}$. Checkable sufficient conditions on the observable and ansatz under which this holds are given in Remark E.1 below.

Step 4: Null space growth for $P > 2E + 1$. For $P > 2E + 1$, $\dim(\ker J^{(\text{par})}) > 0$ by Step 2, so $\sigma_{\min}(J^{(\text{par})}) = 0$ and parameters begin accumulating in $\ker J^{(\text{par})}$. By contrast, for serial circuits $\dim(\ker J^{(\text{ser})})/P \to 1$ across the same range, since phase-locking confines all columns to $\mathcal{T}_L$ regardless of P. ☐

Remark E.1 (Checkable conditions for $\sigma_{\min}(J^{(\text{par})}) > 0$)**.** The real-analyticity argument (Appendix E, Step 3) establishes $\sigma_{\min}(J^{(\text{par})}) > 0$ generically under the following checkable sufficient conditions:

(C1) **Parameter count**: $P \leq 2E + 1$, so the map $\Phi : \mathbb{R}^P \to \mathbb{R}^{2E+1}$ goes from a lower- to higher-dimensional space.

(C2) **Non-trivial observable**: M has at least two distinct eigenvalues (e.g. $M = Z_{N-1}$ as used throughout), so $\langle 0|U^\dagger M U|0\rangle$ is non-constant in $\boldsymbol{\theta}$.

(C3) **Circuit expressivity**: the ansatz generates a sufficiently rich Lie algebra. For the Rot-CNOT ladder used in our experiments, the generators span $\mathfrak{su}(2^N)$ (universal gate set (Brylinski & Brylinski, 2002)), so Φ is non-constant on every open set and $\det(J^{(\text{par})\top} J^{(\text{par})})$ is not identically zero.

Under (C1)–(C3), the zero set of $q(\boldsymbol{\theta}) = \det(J^{(\mathrm{par})\top} J^{(\mathrm{par})})$ has Lebesgue measure zero by real-analyticity (Bochnak et al., 1998, Prop. 2.2.7), so $\sigma_{\min}(J^{(\mathrm{par})}) > 0$ for almost all $\boldsymbol{\theta} \in \mathbb{R}^P$. Degenerate exceptions (where $\sigma_{\min} = 0$ despite $P \leq 2E + 1$) include: $M \propto I$ (trivial output), circuits with a continuous symmetry fixing all Fourier coefficients, or initializations in a measure-zero symmetry-protected submanifold. For the Rot-CNOT ladder with $M = Z_{N-1}$, none of these apply generically.

F Serial-Parallel Separation Theorem

Theorem F.1 (Serial-parallel conditioning separation)**.** *Fix encoding budget $E = NL$ and equal parameter distribution $P_n = P/N$ for the parallel case.*

(i) ***Serial*** *($N = 1$, $L = E$): $\mathrm{rank}(J^{(\mathrm{ser})}) \leq 2L + 1$ independently of P, and $\dim(\ker J^{(\mathrm{ser})})/P \to 1$ as $P \to \infty$ at fixed L: an ever-growing fraction of parameters receive identically zero gradient signal.*

(ii) ***Parallel product ansatz*** *(N qubits, $L = E/N$, no inter-qubit entanglement, equal $P_n = P/N$): for $P \leq 2E + 1$, $\sigma_{\min}(J^{(\mathrm{par})}) > 0$ generically and no parameter lies in $\ker J^{(\mathrm{par})}$; for $P > 2E + 1$, null directions appear and $\dim(\ker J^{(\mathrm{par})})/P \to 1$ as $P \to \infty$.*

For entangling ansätze, the phase-locking argument extends qualitatively; the mechanistic justification is given in Remark 3.1.

Proof. Claim (i) is Proposition 3.3. Claim (ii), $P \leq 2E + 1$: by the absence of phase-locking established in Appendix E (Step 1) and the real-analyticity argument of Step 3, $\sigma_{\min}(J^{(\mathrm{par})}) > 0$ generically, so no parameter lies in $\ker J^{(\mathrm{par})}$. Claim (ii), $P > 2E + 1$: by rank-nullity applied to the shared spectrum $|\Omega| = 2E + 1$, $\dim(\ker J^{(\mathrm{par})}) \geq P - (2E + 1) > 0$, so $\sigma_{\min}(J^{(\mathrm{par})}) = 0$ and null directions appear at the same threshold as the serial case. See Appendix E for details. □

G Interpolation Threshold Table

Table 1: Interpolation threshold vs. R^2 saturation in the tbl sweep ($L = L_{\min}$ fixed, $n_{\mathrm{train}} = 200$). $\mathrm{tbl}_{\mathrm{interp}}$: first tbl where $P \geq n_{\mathrm{train}}$. Gap $= \mathrm{tbl}_{R^2} - \mathrm{tbl}_{\mathrm{interp}}$. Negative gaps indicate saturation before the interpolation threshold; zero gap indicates coincidence; positive gaps indicate that additional parameters beyond the interpolation threshold are needed before gradient directions are sufficiently well-covered.

N	deg	$L_{\min}$	$P(\mathrm{tbl}{=}1)$	$\mathrm{tbl}_{\mathrm{interp}}$	$\mathrm{tbl}_{R^2 \geq 0.95}$	Gap
1	12	12	39	6	3	-3
2	20	10	66	4	4	0
4	28	7	96	3	6	$+3$
6	6	1	36	6	4	-2

The $N = 1$ gap of -3 is noteworthy: saturation occurs *before* the interpolation threshold because at $L_{\min} = 12$, the $2L + 1 = 25$ effective Jacobian directions are already well-covered by $P = 117$ parameters at tbl$= 3$, so gradient-based optimization succeeds before the classical interpolation threshold is reached. The $N = 6$ gap of -2 reflects a similar effect: with $L_{\min} = 1$ and a degree-6 target well within the circuit's accessible spectrum, the $2NL_{\min} + 1 = 13$ Fourier directions are already well-matched at $P = 144$ (tbl$^* = 4$), again before the classical interpolation threshold $P \geq 200$ is reached. By contrast, the positive gap for $N = 4$ ($+3$) indicates that additional parameters beyond the interpolation threshold are needed before the optimizer can sufficiently cover the available gradient directions.

H Parameter Efficiency Table

Table 2 quantifies the parameter efficiency of the FM route vs. the trainable blocks route to reach mean $R^2 \geq 0.95$, starting from $P_{\mathrm{base}} = (L_{\min} + 1) \cdot N \cdot 3$.

Table 2: Parameter efficiency of the FM route vs. the trainable blocks route to reach mean $R^2 \geq 0.95$, starting from $P_{\mathrm{base}} = (L_{\min} + 1) \cdot N \cdot 3$. L^*: smallest L achieving mean $R^2 \geq 0.95$ along the FM route (tbl$= 1$ fixed). tbl*: smallest tbl achieving mean $R^2 \geq 0.95$ along the trainable blocks route ($L = L_{\min}$ fixed). Ratio $= P_{\mathrm{tbl}}/P_{\mathrm{FM}}$.

N	deg	P_{base}	L^*	P_{FM}	tbl*	P_{tbl}	Ratio
1	12	39	20	63	3	117	1.9×
2	20	66	19	120	4	264	2.2×
4	28	96	22	276	6	576	2.1×
6	6	36	4	90	4	144	1.6×

I Experimental Details

Circuit architecture. All experiments use the Rot ansatz with CNOT ladder entanglement: each trainable block consists of a $\mathrm{Rot}(\phi, \theta, \omega)$ gate on wire 0 followed by alternating CNOT and Rot gates on wires $1, \dots, N-1$. Data encoding uses unary $\mathrm{RX}(x)$ gates on all qubits. The observable is $M = Z_{N-1}$, the Pauli-Z operator on the final qubit. All circuits are implemented in PennyLane (Bergholm et al., 2018) with a JAX backend (Bradbury et al., 2018).

Optimizer and initialization. All models are trained with Adam (Kingma & Ba, 2014) at learning rate $\eta = 10^{-3}$ for 5,000 steps. Parameters are initialized uniformly in $[0, 2\pi]$, matching standard practice for variational quantum circuits (Schuld et al., 2021). Each experiment is repeated over 10 random seeds per target function and 5 target functions, giving 50 independent runs per architecture and degree.

Target functions. Synthetic targets are random Fourier series of fixed degree d:

$$f^*(x) = a_0 + \sum_{k=1}^{d} \left[a_k \cos(kx) + b_k \sin(kx)\right], \tag{34}$$

with coefficients $a_k, b_k \sim \mathcal{N}(0, 1)$ and f^* normalized to unit variance. Training uses $n_{\text{train}} = 200$ points drawn uniformly from $[0, 2\pi]$; evaluation uses $n_{\text{test}} = 100$ held-out points.

Degree selection. The target degree for each architecture is chosen as the smallest degree at which the minimal configuration ($L = L_{\min}$, tbl$=1$) fails to reach $R^2 \geq 0.95$: deg$=12$ for $N = 1$, deg$=20$ for $N = 2$, deg$=28$ for $N = 4$, and deg$=6$ for $N = 6$. This places each experiment in a regime where the starting configuration is genuinely insufficient and adding parameters along either route is necessary to reach saturation.

Jacobian diagnostics. The coefficient matching Jacobian $J \in \mathbb{R}^{(2E+1)\times P}$ is computed by differentiating through a Fourier coefficient extraction function: for each architecture (N, L), the circuit output is evaluated on a DFT grid of $n_{\text{diag}} = 2N(L+1)+1$ equally spaced points $x_k = 2\pi k / n_{\text{diag}}$, $k = 0, \dots, n_{\text{diag}} - 1$, and the real Fourier representatives $\tilde{\mathbf{c}}(\boldsymbol{\theta})$ (Definition 2.3) are extracted via FFT. The Jacobian $J = \partial\tilde{\mathbf{c}}/\partial\boldsymbol{\theta}$ is then computed using `jax.jacrev`. The grid size $n_{\text{diag}} = 2N(L+1)+1$ slightly exceeds the theoretical spectrum size $2E+1 = 2NL+1$; the $2N$ additional DFT points lie outside the circuit's accessible spectrum and produce near-zero Jacobian rows, so the rank is bounded by $\min(2E+1, P)$ with no artificial truncation. Singular values are computed via `jnp.linalg.svd`; hard rank counts all singular values above a threshold of 10^{-10}. The Jacobian QFIM $\hat{\mathcal{F}} = \mathbf{J}_{\text{data}}^\top \mathbf{J}_{\text{data}}/n$ used as the spectral diagnostic in Section 4 is computed analogously at convergence for the FM vs. trainable blocks experiments.

Phase-locking experiment (Figure 2). To validate Lemma 3.2 directly, we compute the gradient coefficient $g_j(x) = \langle \psi^\perp | \partial_j \psi \rangle \in \mathbb{C}$ as x sweeps $[0, 2\pi]$ at a fixed random initialization, for a serial circuit ($N = 1$, $L = 4$, $P = 15$) and a parallel circuit ($N = 2$, $L = 2$, $P = 18$). For each parameter j, $g_j(x)$ is computed via `jax.jacrev` applied to the complex circuit output before taking the expectation value, normalized by $\max_x |g_j(x)|$, and plotted as a parametric curve in $\mathbb{C}$. Phase-locking is quantified as $\mathrm{Im}(g_j/g_k)$ averaged over x for all pairs (j, k); a value of 0 indicates perfect phase-locking (Lemma 3.2), while nonzero values indicate independent phase trajectories.

Rank ceiling experiment (Figure 5). To validate Proposition 3.3(i), we sweep encoding budget $E \in \{2, 4, 6, 8, 10, 12, 16\}$ across architectures $N \in \{1, 2, 4\}$ at matched parameter counts $P \approx 3E$. For each (E, N) pair, 100 random initializations are evaluated before training. The coefficient matching Jacobian $J \in \mathbb{R}^{(2E+1)\times P}$ is computed on a DFT grid of $n_{\text{diag}} = 2N(L+1)+1$ equally spaced points (see Jacobian diagnostics paragraph above), with $\mathrm{rank}(J)$ computed from the resulting singular values before training. The serial architecture uses $N = 1$, $L = E$; parallel architectures use $L = E/N$ FM layers per qubit. For $N = 4$, only E divisible by 4 are used to ensure integer L; non-integer rounding would reduce the actual encoding budget and produce an artifactual gap below the ceiling.

FM vs. trainable blocks experiment (Figure 3). To validate the efficiency claims of Section 4, we run two sweeps for each architecture: the FM route sweeps L from $L_{\min}$ to $L_{\max}$ with $\mathrm{tbl} = 1$ fixed, and the trainable blocks route sweeps tbl from 1 to $\mathrm{tbl}_{\max}$ with $L = L_{\min}$ fixed. The ranges are:

N	$L_{\min}$	$L_{\max}$	$\mathrm{tbl}_{\max}$
1	12	89	10
2	10	40	4
4	7	32	8
6	1	21	10

$L_{\max}$ is chosen to extend well beyond the saturation threshold L^*; $\mathrm{tbl}_{\max}$ is chosen to include at least one configuration beyond tbl*. Both sweeps use $N \in \{1, 6\}$ in the main text and $N \in \{2, 4\}$ in Appendix O. The Jacobian QFIM is evaluated at convergence, averaged over all runs, to produce the normalized eigenvalue spectra of Figure 3.

Real-world validation experiments (Figures 7 and 8). Both experiments use the Nottingham temperature dataset (Hipel & McLeod, 1994) with $n_{\mathrm{train}} = 200$ and $n_{\mathrm{test}} = 40$ points by fixed random split, time index mapped to $[0, 2\pi]$, target scaled to $[-1, 1]$. *Experiment 1* ($E = N \times L_{\min} = 12$): $L_{\min} \in \{12, 6, 3, 2\}$ for $N \in \{1, 2, 4, 6\}$. *Experiment 2* ($E = N \times L_{\min} = 24$): $L_{\min} \in \{24, 12, 6, 4\}$ for $N \in \{1, 2, 4, 6\}$. For both experiments, FM sweep step sizes are 6, 3, 2, 1 for $N = 1, 2, 4, 6$ respectively (approximately constant parameter increment $\sim$18 per step); the tbl sweep uses tbl$= 1, \ldots, 20$ with $L = L_{\min}$ fixed. Optimizer, initialization, and Jacobian diagnostic settings are identical to the synthetic experiments.

J Ruling Out Barren Plateaus

Barren plateaus (McClean et al., 2018) are characterized by exponential decay of gradient variance with circuit depth. Figure 4 shows that gradient variance is approximately constant within each architecture as parameter count increases, with no sign of this decay. Crucially, the serial architecture $N = 1$ has the largest gradient variance ($\approx 10^{-2}$) yet fails to train (cf. Figure 1), directly ruling out gradient vanishing as the primary failure mode.

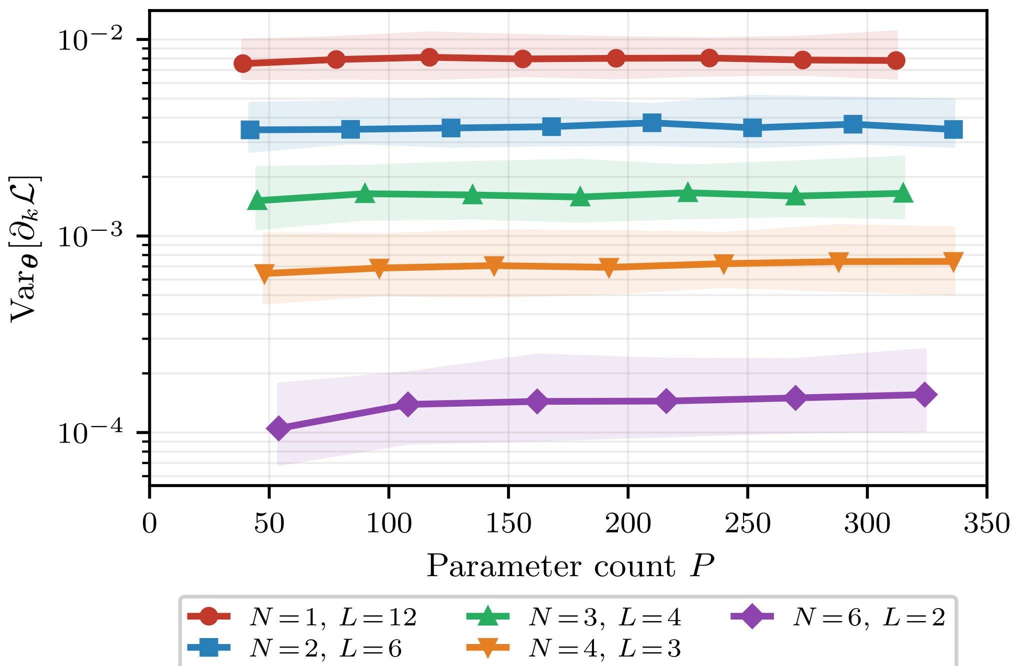


Figure 4: Gradient variance $\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k \mathcal{L}]$ vs. parameter count p for degree-12 targets, separated by architecture. Median over random initializations; shaded bands show IQR. Variance is approximately constant within each architecture, with no sign of depth-dependent exponential decay.

K Jacobian Rank Ceiling: Empirical Validation

Figure 5 validates Proposition 3.3(i) empirically across encoding budgets $E \in \{2, 4, 6, 8, 10, 12, 16\}$ and architectures $N \in \{1, 2, 4\}$ at matched parameter counts $P \approx 3E$. All architectures reach the rank ceiling $2E + 1$ at matched P, confirming that the ceiling is tight: at $P > 2E + 1$, a growing fraction $(P - (2E + 1))/P$ of parameters lie in $\ker J$ regardless of architecture. For $N = 4$, only E divisible by 4 are used to ensure integer $L = E/N$; non-integer rounding would silently reduce the actual encoding budget and produce an artifactual gap below the ceiling.

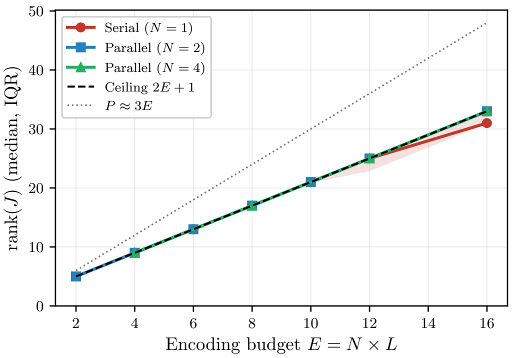
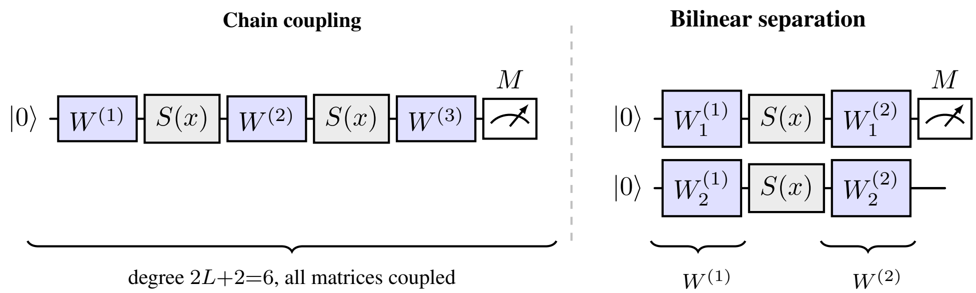


Figure 5: **Jacobian rank saturates at the $2E + 1$ ceiling for all architectures at matched $P \approx 3E$.** Median $\mathrm{rank}(J)$ with IQR across 100 random initializations at matched parameter counts $P \approx 3E$ (dotted gray). All architectures track the theoretical ceiling $2E + 1$ (dashed black), confirming Proposition 3.3(i): the ceiling is tight and at $P > 2E + 1$ a fraction $(P - (2E + 1))/P$ of parameters lie in $\ker J$ regardless of architecture. For $N = 4$, only E divisible by 4 are shown to ensure integer $L = E/N$.

L Bilinear Factorization: Circuit Diagram

Figure 6 illustrates the structural difference between serial and parallel coefficient matching equations at encoding budget $E = 2$, complementing the algebraic comparison of Section 3.1. The contrast between chain coupling (serial) and bilinear separation (parallel) is visible directly from the circuit structure: in the serial case all three parameter matrices are coupled through the chain of encoding layers, while in the parallel case $W^{(1)}$ and $W^{(2)}$ enter through entirely separate factors.



$$c_{+2}^{\mathrm{ser}} = -2\,\alpha_1\,\alpha_2^2\,\bar{\beta}_1\,\beta_3\bar{\alpha}_3$$

$$c_{+2}^{\mathrm{par}} = \underbrace{v_1\bar{v}_4}_{W^{(1)}} \cdot \underbrace{\widetilde{M}_{4,1}}_{W^{(2)}}$$

Figure 6: Structural comparison of coefficient matching equations at encoding budget $E = 2$. **Left (serial, $N = 1$, $L = 2$):** c_{+2}^{ser} couples all three trainable parameter matrices simultaneously in a degree-6 polynomial — every gradient update entangles all parameter blocks. **Right (parallel, $N = 2$, $L = 1$):** c_{+2}^{par} factors into a $W^{(1)}$-dependent term and a $W^{(2)}$-dependent term with no cross-coupling (Proposition A.1), consistent with the independent phase trajectories of Corollary 3.3.1. Blue boxes: trainable ansatz layers. Gray boxes: data-encoding layers $S(x)$. CNOT entanglement omitted for clarity; see Remark 3.1.

M Real-World Validation: Nottingham Temperature Dataset

We validate the structural predictions of Sections 3 and 4 on the Nottingham temperature dataset (Hipel & McLeod, 1994), a canonical univariate time series of 240 monthly mean air temperatures recorded at Nottingham Castle, England, from 1920 to 1939. We map the time index to $[0, 2\pi]$, scale the target to $[-1, 1]$, and use $n_{\mathrm{train}} = 200$ and $n_{\mathrm{test}} = 40$ points by fixed random split. Two experiments are reported using different encoding budgets $E = N \times L_{\min}$ to probe different regimes: $E = 12$ (Figure 7) places all architectures in the expressivity-limited regime where the model spectrum is insufficient at $L_{\min}$; $E = 24$ (Figure 8) gives each architecture sufficient expressivity at $L_{\min}$ and isolates the parameter efficiency question.

Experiment 1: $E = N \times L_{\min} = 12$ **(Expressivity Regime)**

With $E = N \times L_{\min} = 12$, the encoding budgets are $L_{\min} \in \{12, 6, 3, 2\}$ for $N \in \{1, 2, 4, 6\}$ respectively. Unlike the synthetic experiments, the Nottingham spectrum is not perfectly bandlimited: significant spectral content lies beyond $L_{\min}$ coverage, so FM layers provide a dual benefit — expanding Jacobian rank coverage *and* expanding expressivity — while the trainable blocks route can only improve conditioning within the fixed rank ceiling without expanding the accessible function class.

The trainable blocks route is stuck at $R^2 < 0$ across the full parameter range for all four architectures. For $N = 2$ and $N = 4$, which successfully trained on synthetic targets via the tbl route, the failure here is not gradient starvation but an expressivity ceiling: the Nottingham spectrum extends beyond the function class accessible at fixed $L_{\min}$, and adding trainable blocks cannot expand it. For $N = 1$, structural gradient starvation compounds the expressivity failure. The FM route succeeds for $N \in \{1, 2, 4\}$ by raising L, simultaneously expanding the accessible spectrum and the Jacobian rank ceiling. For $N = 6$, the FM route improves monotonically but plateaus around $R^2 \approx 0.80$; the structural interpretation is given in Experiment 2 below.

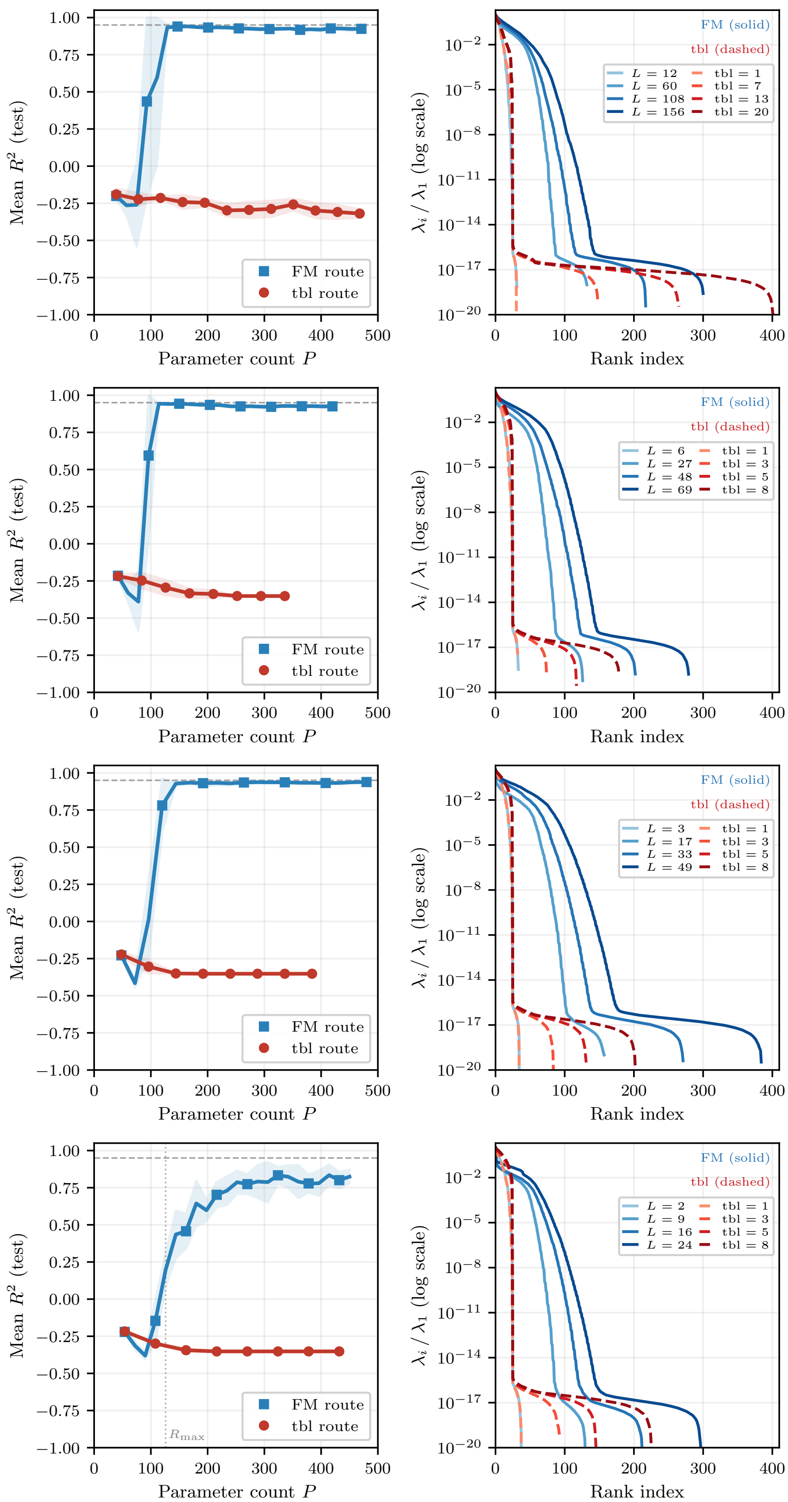


Figure 7: **Real-world validation on the Nottingham temperature dataset** ($E = N \times L_{\text{min}} = 12$, $n_{\text{train}} = 200$)**.** Each row shows one architecture ($N \in \{1, 2, 4, 6\}$). **Left panels:** mean R^2 (test) $\pm$ std vs. parameter count P; dotted vertical line marks $R_{\text{max}} = 126$ for $N = 6$ only. **Right panels:** normalized Jacobian QFIM spectra λ_i/λ_1 at selected L (FM, solid blue) and tbl (trainable blocks, dashed red) values. The trainable blocks route fails categorically across all four architectures ($R^2 < 0$ throughout), while the FM route reaches $R^2 \geq 0.95$ for $N \in \{1, 2, 4\}$ and approaches but does not fully reach saturation for $N = 6$. The spectral knee shifts rightward along the FM route and remains frozen along the tbl route in all four cases, consistent with Figure 3.

Experiment 2: $E = N \times L_{\min} = 24$ **(Parameter Efficiency Regime)**

With $E = N \times L_{\min} = 24$, the encoding budgets are $L_{\min} \in \{24, 12, 6, 4\}$ for $N \in \{1, 2, 4, 6\}$ respectively, giving each architecture sufficient expressivity at $L_{\min}$ to represent the target spectrum. This isolates the parameter efficiency question from expressivity effects.

FM efficiency for $N \in \{1, 2, 4\}$**.** For $N = 1$, the FM route reaches $R^2 \approx 0.94$ at $L = 43$ ($P = 129$) while the tbl route reaches comparable performance at tbl$= 7$ ($P = 525$), a $4.1\times$ efficiency ratio. For $N = 2$, the FM route reaches $R^2 \geq 0.94$ at $L = 19$ ($P = 114$) vs. tbl$= 4$ ($P = 312$), a $2.7\times$ ratio. For $N = 4$, FM reaches $R^2 \geq 0.93$ at $L = 15$ ($P = 180$) vs. tbl$= 4$ ($P = 336$), a $1.9\times$ ratio. These efficiency ratios are consistent with the synthetic results of Table 2, confirming that the FM parameter efficiency advantage generalizes to real-world data when expressivity is not the binding constraint.

Inversion for $N = 6$ **and the Larocca boundary.** For $N = 6$, the result inverts: the tbl route reaches $R^2 \geq 0.95$ at tbl$= 5$ ($P = 450$) while the FM route plateaus around $R^2 \approx 0.83$ despite each FM layer expanding the accessible spectrum by $N = 6$ encoding units. The right panel for $N = 6$ shows the spectral knee growing less with each FM layer than for $N \in \{1, 2, 4\}$, and eigenvalue mass concentrated at lower rank indices. This is consistent with the $N = 6$ circuit entering the Larocca underparameterization regime at $P \approx R_{\max} = 126$ (Larocca et al., 2023) early in the FM sweep: at $L_{\min} = 4$, $P_{\text{base}} = 90$, and the FM route crosses $R_{\max}$ at $L = 5$ ($P = 108$). Beyond this point, adding FM layers expands the model spectrum and requires control of additional Fourier coefficients, increasing the parameter demand faster than the added parameters satisfy it. The tbl route avoids this by keeping $L = L_{\min} = 4$ fixed and reaching saturation through the classical interpolation mechanism, unaffected by the Larocca condition. Whether this inversion is a general property of the Larocca-limited regime or specific to the $N = 6$ architecture and Nottingham target is left for future investigation.

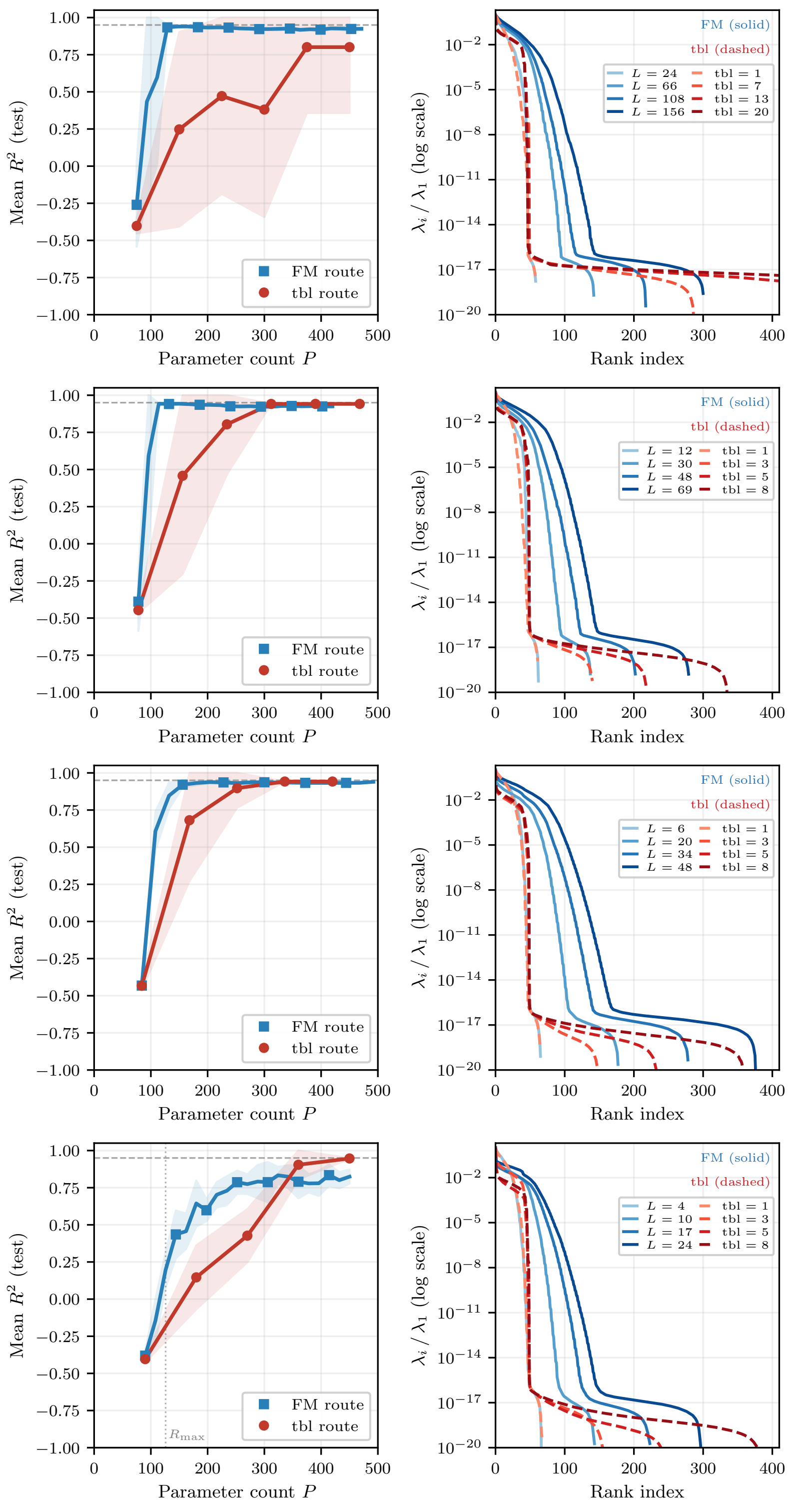


Figure 8: **Real-world validation on the Nottingham temperature dataset ($E = N \times L_{\min} = 24$, $n_{\text{train}} = 200$).** Layout identical to Figure 7. $L_{\min} \in \{24, 12, 6, 4\}$ for $N \in \{1, 2, 4, 6\}$ respectively; $R_{\max} = 126$ line shown for $N = 6$ only. For $N \in \{1, 2, 4\}$, the FM route reaches $R^2 \geq 0.94$ with 1.9–2.7× fewer parameters than the trainable blocks route, consistent with the synthetic results of Table 2. For $N = 6$, the result inverts: the tbl route reaches $R^2 \geq 0.95$ at tbl$=5$ ($P = 450$) while the FM route plateaus around $R^2 \approx 0.83$, consistent with the circuit entering the Larocca underparameterization regime $P \approx R_{\max} = 126$ (Larocca et al., 2023) early in the FM sweep.

N Spectral Knee Position: Quantitative Diagnostic

Figure 9 quantifies the spectral knee position — the number of eigenvalues satisfying $\lambda_i/\lambda_1 > 10^{-6}$ — as a function of parameter count P along both routes for $N = 1$ and $N = 6$.

For $N = 1$ (left panel), the FM route shifts the knee monotonically from rank 25 at $L = L_{\min} = 12$ to rank 52 at $L = 89$: each extra FM layer expands the ceiling $2L + 1$ and makes genuinely new Fourier directions accessible. The trainable blocks route knee is frozen at rank 24–25, matching the theoretical ceiling of 25, with no monotonic growth — confirming that no new Fourier directions become accessible as trainable blocks are added.

For $N = 6$ (right panel), the FM route shifts the spectral knee monotonically from rank 13 at $L = L_{\min} = 1$ to rank 79 at $L = 21$, and both routes reach $R^2 \geq 0.95$. The FM route reaches saturation at $L^* = 4$ ($P = 90$) while the tbl route reaches saturation at tbl$^* = 4$ ($P = 144$), giving a $1.6\times$ efficiency ratio. The tbl knee is frozen at 12–13, matching the theoretical rank ceiling of $2NL_{min} + 1 = 13$ regardless of parameter count, consistent with the frozen-ceiling pattern seen across all architectures: the tbl route improves R^2 via the classical interpolation mechanism rather than by accessing new Fourier directions.

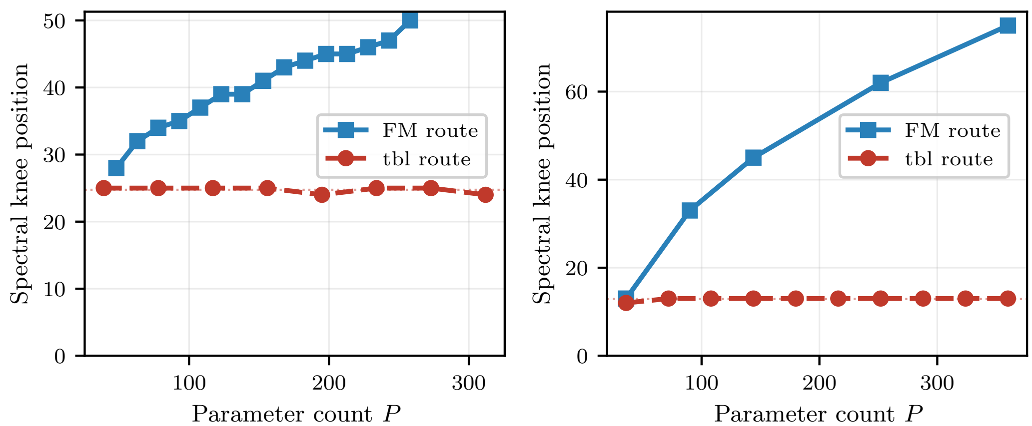


Figure 9: **Spectral knee position vs. parameter count P along the FM route (blue) and trainable blocks route (red).** The knee is defined as the number of normalized eigenvalues λ_i/λ_1 exceeding 10^{-6}, a threshold-free measure stable across several decades. **Left ($N = 1$):** the FM knee grows monotonically from rank 25 at $L_{\min} = 12$ to rank 52 at $L = 89$, while the trainable blocks knee is frozen at rank 24–25, matching the theoretical rank ceiling $2L_{\min} + 1 = 25$, regardless of parameter count — confirming the frozen-ceiling claim of Section 4.2. **Right ($N = 6$):** the FM knee grows from rank 13 to 79 while the tbl knee is frozen at 12–13, matching the theoretical rank ceiling $2NL_{\min} + 1 = 13$; both routes reach $R^2 \geq 0.95$.

O FM vs. Trainable Blocks: Full Results for $N = 2$ and $N = 4$

Figure 10 shows the FM and classical routes for $N = 2$, deg$= 20$ and $N = 4$, deg$= 28$ — the two architectures in the well-conditioned regime where both routes reach mean $R^2 \geq 0.95$ within the tested range. These complement the main text results for $N = 1$ and $N = 6$ (Figure 3), which illustrate contrasting efficiency regimes.

For both $N = 2$ and $N = 4$, the qualitative picture matches the theoretical prediction: the FM route reaches saturation with $2.2\times$ fewer parameters than the trainable blocks route for $N = 2$ and $2.1\times$ for $N = 4$ (Table 2), and the normalized QFIM eigenvalue spectra show the spectral knee shifting rightward along the FM route while remaining fixed along the trainable blocks route. The frozen-knee pattern is particularly clear for $N = 2$, where the tbl curves collapse visibly at the same knee position across tbl$= 1 \ldots 4$ despite a four-fold increase in parameter count. For $N = 4$ the FM curves are compressed near the left edge of the rank axis, reflecting the larger state space ($2^N = 16$) which distributes eigenvalue mass across more directions. The $N = 4$ tbl sweep exhibits non-monotonic R^2 behavior: mean R^2 dips from 0.621 at tbl$= 1$ to 0.570 at tbl$= 2$ before recovering monotonically to 0.984 at tbl$= 6$. The cause of this dip was not investigated further; it does not affect the structural conclusion that the FM route is $2.1\times$ more parameter-efficient.

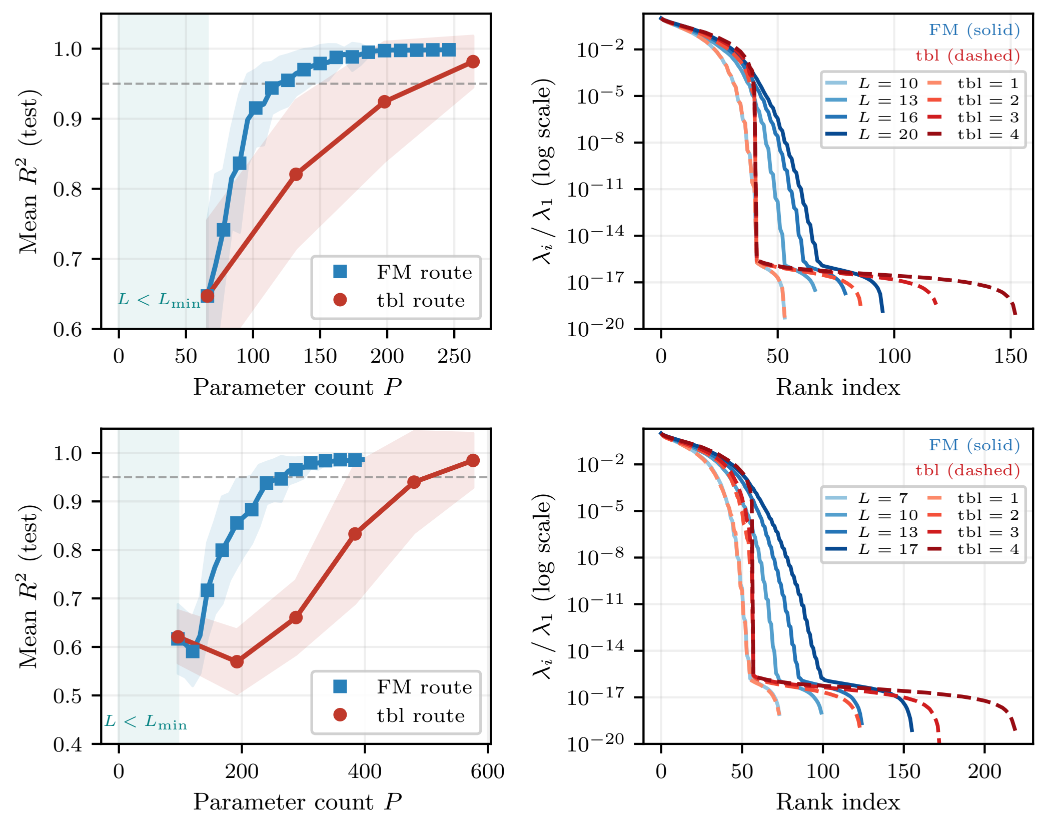


Figure 10: **FM route vs. trainable blocks route for $N = 2$ (top row) and $N = 4$ (bottom row). Left panels:** mean R^2 (test) $\pm$ std vs. parameter count P. Teal shading: expressivity gap ($L < L_{\min}$). Dashed gray line: $R^2 = 0.95$ threshold. Both routes reach saturation; the FM route requires $2.2\times$ fewer parameters for $N = 2$ and $2.1\times$ for $N = 4$ (Table 2). **Right panels:** normalized Jacobian QFIM spectra λ_i/λ_1 at selected L values (FM, solid blue) and tbl values (trainable blocks, dashed red). The spectral knee generally shifts rightward along the FM route and remains fixed along the trainable blocks route, consistent with Figure 3 and the frozen-knee claim of Section 4.2.

P Degree Selection for Efficiency Experiments

The efficiency comparison in Table 2 uses a fixed target Fourier degree for each architecture: deg$= 12$ for $N = 1$, deg$= 20$ for $N = 2$, deg$= 28$ for $N = 4$, and deg$= 6$ for $N = 6$. Figure 11 justifies these choices by showing mean R^2 (test) as a function of target degree for each architecture, evaluated at matched encoding budget $E = \deg$ with tbl$= 1$.

For $N = 2$ and $N = 4$, the chosen degrees sit at the onset of reliable degradation: mean R^2 remains near 1.0 for lower degrees and begins to decline consistently beyond the chosen value, placing the experiment in a regime where the architecture is genuinely challenged without already failing. For $N = 1$, deg$= 12$ falls within the declining regime — consistent with the gradient starvation mechanism, which causes serial circuits to degrade at lower degrees than parallel architectures of the same encoding budget. For $N = 6$, deg$= 6$ is the lowest tested degree; the minimum configuration ($L_{\min} = 1$, tbl$= 1$, $P = 36$) lies well below the geometric threshold $R_{\max} \leq 2^{N+1} - 2 = 126$ of Larocca et al. (2023), explaining the low $R^2 \approx 0.56$ at the starting point. Both routes reach $R^2 \geq 0.95$ once P approaches or exceeds $R_{\max}$: the tbl route crosses the threshold around tbl$= 4$ ($P = 144$) and the FM route at $L^* = 4$ ($P = 90$), where the accessible spectrum ($2NL + 1 = 49$) remains well within $R_{\max} = 126$. The dashed vertical lines mark the chosen degree for each architecture.

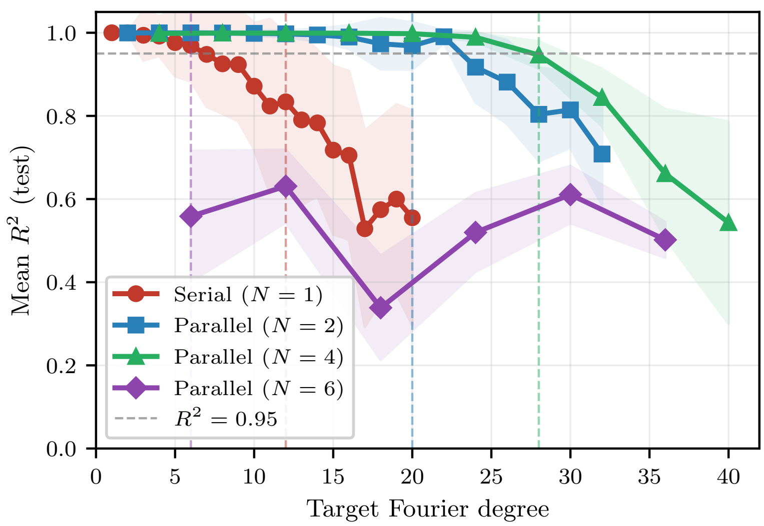


Figure 11: Mean R^2 (test, $\pm$ std) vs. target Fourier degree for each architecture, at matched encoding budget $E = \deg$ and tbl$= 1$. Dashed vertical lines mark the degree used in Table 2 for each architecture. Dashed gray line: $R^2 = 0.95$ reference. For $N = 2$ and $N = 4$, the chosen degree sits at the onset of degradation. For $N = 1$, deg$= 12$ falls within the declining regime, consistent with structural gradient starvation. For $N = 6$, the minimum configuration lies below the geometric threshold $R_{\max} = 126$ (Larocca et al., 2023); both routes reach $R^2 \geq 0.95$ once P approaches $R_{\max}$.